MILITARY SPECIFICATION

RECEIVER-TRANSMITTER, RADIO RT-1324A/ARC, RT-1327A/ARC and RT-1407/ARC

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for Radio RT-1324A/ARC, RT-1327A/ARC and RT-1407/ARC Receiver-Transmitters hereinafter referred to as receiver-transmitter. The receiver-transmitter is capable of providing simplex two-way communication of normal and secure voice amplitude modulated (AM) or frequency modulated (FM) signals.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-S-25879	Switch, Coaxial, Radio Frequency Transmission Line
MIL-M-38510	Microcircuits General Specification for
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environmental Resistant, Removable Crimp Contacts; General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-P-55110	Printed Wiring Boards

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IML-C-85666	Equipment Specification, Radio Control Units C-10319A/ARC, C-10776A/ARC, and C-11131/ARC
MIL-C-85806	Equipment Specification, Control, Radio Set, C-11128/ARC, C-11129/ARC and C-11130/ARC

STANDARDS

Federal

FED-STD-595	Federal Standard Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Electronic Equipment, General Requirements for
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Aircraft, Electric Power, Characteristics and Utilization of
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for

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MIL-STD-2084	Maintainability of Avionics and Electronic Systems and Equipment, General Requirements for

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

Naval Air Systems Command

ADS-23	Aircrew Station Lighting for Compatibility with Night Vision Goggle Use
EI-669	Avionics Installation Instructions for Radio Set AN/ARC-182

National Security Agency

CSEEB-13	TSEC/KY-28 Speech Security Equipment Engineering Bulletin
CSESD-14	Communications Security Equipment System Document for TSEC/KY-57/58
CSEEB-32	TSEC/KY-57/58 Communications Security Equipment Engineering Bulletin

Department of the Navy

NAVMAT P-4855.1	Navy Power Supply Reliability
NAVMAT P-9492	Navy Manufacturing Screening Programs

Naval Air Systems Command

WS-6536	Process Specification Procedures and Requirements for Preparations and Soldering of Electrical Connections

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item description. The receiver-transmitter shall be capable of providing simplex two-way communication of normal and secure voice amplitude modulated (AM) or frequency modulated (FM) signals in 25 Kilohertz (KHz) increments within RF bands: 30.000 MHz through 87.975 MHz (FM), 118.000 MHz through 155.975 MHz (AM), 156.000 MHz through 173.975 MHz (FM) and 225.000 through 399.975 MHz (AM and FM). The receiver-transmitter shall provide a receive only mode of operation of normal and secure voice from 108.000 MHz through 117.975 MHz (AM).

3.2 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Parts. In the selection of parts, the prime considerations are fulfillment of performance requirements, logistics supportability and reliability assurance in a cost effective design. The following shall govern in parts selection.

a. Subassemblies shall be designed and fabricated to be repairable. Maximum economic standardization of parts shall be exercised. Parts selection shall be made in the following order of precedence:

(1) Military Standard parts shall be selected in accordance with MIL-STD-454, Requirement 22.

(2) Nonstandard parts previously qualified for other programs.

(3) Nonstandard parts.

b. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the requirements of the most similiar military standard or standard nonestablished reliability part. Integrated circuits and hybrids shall use MIL-STD-883, Method 5004, Class B, as a basis for tailored screens.

3.3.1 Component rescreening. The contractor shall conduct 100 percent incoming inspection of all active components. Functional and parametric checks of active components shall be performed at the components rated temperature extremes and at ambient. Components which are not compliant with their specification shall not be used in the ARC-182.

3.3.2 Parts derating and application. All parts used shall be applied well within their ratings. The derating shall encompass the appropriate and meaningful application conditions such as voltage, current, power, temperature, mechanical, and duty cycle. Electronic and electromechanical parts shall conform to Table I electronic parts derating for worst case electrical and environmental stress unless formal written approval is received from the procuring activity prior to incorporation into the design (see 6.2.2). Part level stress analysis shall be used to verify that all parameter stresses are within the derated values at worst case circuit and environmental conditions. The following limitations on parts usage shall apply:

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TABLE I. Electronic parts derating.

Part Type	Parameter	Maximum Allowable Application Stress
Capacitor	Applied Voltage (Ripple Voltage for Tantalum)	50% of rating 1/
Resistors	Power Voltage	50% of rating 70% of rating
Diodes	Power Current Voltage Junction Temperature	50% of rating 75% of rating 60% of rating 2/
Transistors	Power Voltage Junction Temperature	55% of rating 60% of rating 2/
Integrated Circuits		
Digital	Fan-Out, Fan-In Junction Temperature	80% of rating 2/
Linear	Power Junction Temperature	75% of rating 2/
Transformers, Inductive Devices	Parts shall be used at a winding temperature of at least 30 degrees below rated temperature. In addition, the maximum voltage between windings or between a winding and a shield shall be not greater than 60% of rated voltage for the steady state conditions and 90% of rated voltage with transients. General purpose inductors, audio and power transformers shall be applied at no more than 70% of rated current.	

1/ For solid tantalum and all electrolytic capacitors, the maximum applied voltage shall be not greater than 70 percent of the vendor rated value. "Applied voltage" is the sum of the applied peak ripple voltage and the applied DC voltage.

2/ Semiconductor devices shall be mounted and operated in such manner to insure worst-case junction temperature is not greater than 110 degrees C at +71 degrees C ambient temperature. Any deviation requires procuring activity approval. Power semiconductor devices, on a very selective basis, may be mounted and operated with worst case junction temperatures not greater than 150 degrees C at +71 degrees C ambient temperatures.

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a. Prohibited application

(1) Non-hermetically sealed semiconductors except power transistors
(2) Photo couplers (photo transistors)
(3) Hot carrier (Schottky) rectifiers
(4) Microcircuit sockets
(5) Non-hermetically sealed wet tantalum capacitors

3.3.3 Parts control program. The contractor shall establish and maintain a parts control program in accordance with the requirements of MIL-STD-965, Procedure I and as specified herein. The program shall control part standardization and the criteria for the selection, application, and testing of parts to achieve the reliability required. As a minimum, a parts program equivalent to that specified herein shall be required of subcontractors and suppliers.

3.3.4 Non-standard parts approval. The contractor shall obtain procuring activity approval of all non-standard parts used in the equipment. Standard parts are defined in MIL-E-5400. In addition to the MIL-E-5400 requirements for standard parts, microelectronic parts are standard only if:

a. The device is listed in MIL-STD-1562, and

b. Sources (at least two) are listed on the Qualified Parts List (QPL), and

c. The part is procured from one of the qualified sources, and

d. Quality level "B" (or better) of MIL-M-38510 is required.

3.3.4.1 Approval procedure. The approval procedure shall be in accordance with MIL-E-5400 and MIL-STD-965, Procedure I. In addition to the requirements of MIL-E-5400, approval of non-standard parts will be based on:

a. Suitability for the applications

b. Conformance to this specification and MIL-E-5400

c. Interchangeability

d. Screening (burn-in) method

3.3.5 Samples of nonstandard parts. Samples of nonstandard parts are not required unless specified in the contract. Where non-standard parts samples are specified in the contract, the sample shall be in accordance with MIL-E-5400 (see 6.2.1).

3.3.6 Cabling and connections.

3.3.6.1 Cables and connectors. The receiver-transmitter shall provide for the use of cables and connectors in accordance with MIL-E-5400. The design of the connectors shall be such that improper assembly, mating or installation is impossible. Non-RF connectors shall meet the requirements of

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MIL-C-38999. All external connectors shall be scoop-proof, have cadmium plating finish in accordance with QQ-P-416 (over a suitable underplating) to withstand a 500 hour salt test, and provide 360 degrees circumferential grounding techniques for grounding prior to contact engagement.

3.3.6.2 Interconnection cabling. The receiver-transmitter shall be capable of required operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the receiver-transmitter.

3.3.7 Control panels. The receiver-transmitter configuration including the front panel controls shall conform to the applicable requirements of MIL-C-6781.

3.3.7.1 Lighted panels-console mounted equipment. The plastic panels for all console mounted receiver-transmitter shall conform to MIL-P-7788, Class 1-R, 1-W or 1-BW Type V, with both 5 and 28-volt (ac and dc) lamp systems installed. Separate input power pins shall be provided for the two lighting systems. Color of panel lighting shall be in accordance with MIL-C-25050 as selected from the options by nomenclature assignment or by description on purchase orders listed in Table II.

TABLE II. Panel lighting color options

Nomenclature Assignment	Color (Short Name)	Color Per MIL-C-25050	MIL-P-7788 Class
RT-1324A	IPL Red	Type 1 (f)	1-R
RT-1327A	IPL White	Type 1 (g)	1-W
RT-1407	Blue Filtered White	Type 1 (h)	1-BW

3.3.7.2 Console mounted equipment function marking. Console mounted control panels shall be permanently and clearly marked to indicate the function of the equipment with which they are associated. Console mounted control panels shall be identified by the abbreviation VHF/UHF located on the front panel.

3.3.7.3 Control knobs and functions. The number of controls required to control the equipment shall be limited to and consistent with performing all the functions required herein. Closely associated controls such as control knobs on concentric shafts, shall be easily identified by contrasting colors of black and grey.

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3.3.7.4 Interchangeability. Physical and functional interchangeability shall exist for all units and replaceable assemblies, subassemblies, and parts of the receiver-transmitter in accordance with MIL-E-5400. The receiver-transmitter shall be constructed to preclude the requirement for any adjustment when replacing an item by another item of the same type designation. Provision shall be made for adjustment of sidetone level, microphone input sensitivity, and squelch thresholds on the main and guard receiver.

3.4 Design and construction. The receiver-transmitter shall conform with all the applicable requirements of MIL-E-5400 for design and construction except as otherwise specified herein. In any case, it is a condition of final acceptance that the receiver-transmitter shall meet all the performance requirements of this specification.

3.4.1 Reliability. Unless otherwise specified in the contract or purchase order, the contractor shall establish a reliability program conforming to MIL-STD-785 (see 6.2.1). Task 104 of MIL-STD-785 shall be implemented to establish and maintain a closed-loop failure reporting system. This system will identify failure trends and patterns at all levels of assembly prior to acceptance of the hardware and will also identify failure trends related to field failures throughout the contract period. The contractor shall develop and implement corrective action for significant failure trends. The reliability program shall include provisions for assuring the reliability of the receiver-transmitter or any portion of the equipment when supplied for use as spare or repair part.

3.4.1.1 Operational stability. The receiver-transmitter shall operate with specified performance for at least the specified mean (operating) time between failures without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The receiver-transmitter shall have a total operating life of 20,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.4.1.3 Reliability in MTBF (mean time between failure). The receiver-transmitter, including, any built-in-test (BIT) provisions, shall have a specified mean (operating) time between failures, upper test MTBF (θ_0) of 1000 hours (as specified in MIL-STD-781), and a lower test MTBF (θ_1) of 500 hours when tested and accepted as specified under the requirements of 4.4.6 and 4.5.4.

3.4.2 Maintainability. Maintainability (built-in test features, construction and packaging, provision for test points and other maintainability parameters) shall be as specified in MIL-STD-2084. The encapsulation requirements of MIL-STD-2084 shall not be required for the three voltage controlled oscillators (VCO), the temperature compensated crystal oscillator (TCXO), the two loop filters, the power amplifier, and the preamplifier assemblies. The fault detection percentage requirement of MIL-STD-2084 shall be not less than 95 percent of the critical path circuitry.

3.4.2.1 Compatibility with ATE. The receiver-transmitter shall be compatible with Automatic Test Equipment (ATE) for WRA testing and fault isolation to a faulty SRA in accordance with MIL-STD-2076.

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3.4.2.2 Maintainability design. All electrical or electronic circuits and parts shall be packaged on replaceable and repairable, plug-in modules. Circuit breakers, indicators, and other items which may logically be bolted and soldered, or otherwise more rigidly electrically and mechanically fastened to the receiver-transmitter housing shall not be considered modules. SRAs (Shop Replacement Assemblies) shall be QRAs (Quick Replaceable Assemblies) as specified in MIL-STD-2076. SRA arrangement shall be such that access to any SRA does not require the removal of adjacent SRAs or parts other than access panels. The use of a special tool or tools shall not be required for SRA removal.

3.4.2.2.1 Keying. Connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the receiver-transmitter. All connectors shall be legibly marked with reference designations.

3.4.2.2.2 Access panels. Access panels shall be retained by fasteners suitable to the procuring activity.

3.4.2.2.3 Service access. Fasteners required to be loosened for removal of the WRA shall be held by retaining devices suitable to the procuring activity.

3.4.2.2.4 Adjustments. No adjustment or alignment shall be required at the organizational level. Adjustment or alignment if required at a higher maintenance level (intermediate or depot), shall not require the removal of the part to be adjusted or aligned, or of the hardware element to which it is attached.

3.4.2.2.5 Test points. All test points shall be in accordance with MIL-STD-2084 and MIL-STD-415 to accommodate approved test equipment. MIL-STD-2084 takes precedence over MIL-STD-415 requirements. The sub-SRA test points and related connector requirements of MIL-STD-2084 are waived where prohibited by space or circuit performance considerations. The sub-SRA test points and related connector requirements philosophy shall be reflected in appropriate maintenance data. Conformal coating may be applied to the PC pads used as test points. Injection test points in the BIT circuits are not required.

3.4.2.2.5.1 Module test points. Module test points shall be accessible while power remains applied to the modules. Test points shall become accessible with minimum removal of protective covers or enclosures.

3.4.2.2.5.2 Test points utilized at intermediate or depot level. Test points shall be provided for verification of faulty equipments and isolation of faulty modules at the intermediate level and repair of equipments and modules at the depot level.

3.4.2.2.5.2.1 Power supply test points. Accessible, individual test points shall be provided on the receiver-transmitter to measure the power supply voltages. The power supply test points and related connector requirements philosophy shall be reflected in appropriate maintenance data.

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3.4.2.2.6 Organizational level maintainability. The receiver-transmitter shall provide the operator or organizational maintenance techician, while airborne or on the ground with performance/readiness test and fault (GO/NO-GO) indication.

3.4.2.2.6.1 Performance or readiness test requirements. BIT provisions shall be incorporated in the receiver-transmitter to provide a GO/NO-GO indication of receiver-transmitter readiness. These provisions may be in conjunction with, but not necessarily as part of, the fault isolation features. The operation of the performance or readiness test provision shall be both a manually energized or continuous test mode of operation. RF radiation is not permitted during performance readiness tests. Normal equipment performance may be interrupted during the manually initiated BIT sequence and, if necessary, switches may be used in series paths for introducing BIT stimuli or measuring performance. Self-test of BIT is not required.

3.4.2.2.6.2 Fault isolation. In conjunction with, but not necessarily as part of, the performance or readiness test features, there shall be features incorporated which provide fault isolation to the SRA.

3.4.2.2.6.3 MTTR (mean time to repair). For organizational maintenance, the MTTR shall be not greater than 15 minutes to remove and replace and verify operation of the replaced equipment. The receiver-transmitter design shall be such that no ancillary test equipment is required for organizational level maintenance.

3.4.2.2.7 Intermediate level maintainability. Intermediate level maintenance may be performed with the assistance of separate test equipment to isolate a fault to the SRA level. The MTTR to isolate the fault to a SRA, replace the SRA, and return to full performance at the intermediate level maintenance shall be not greater than 20 minutes.

3.4.3 Transportability. The receiver-transmitter shall be transportable by air, rail, truck and ship when packaged as specified in Section 5.

3.4.4 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests, except that required field tests may be conducted under outside ambient conditions:

a.	Temperature	Room ambient
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input power voltage	28 $\pm$ 0.5 VDC

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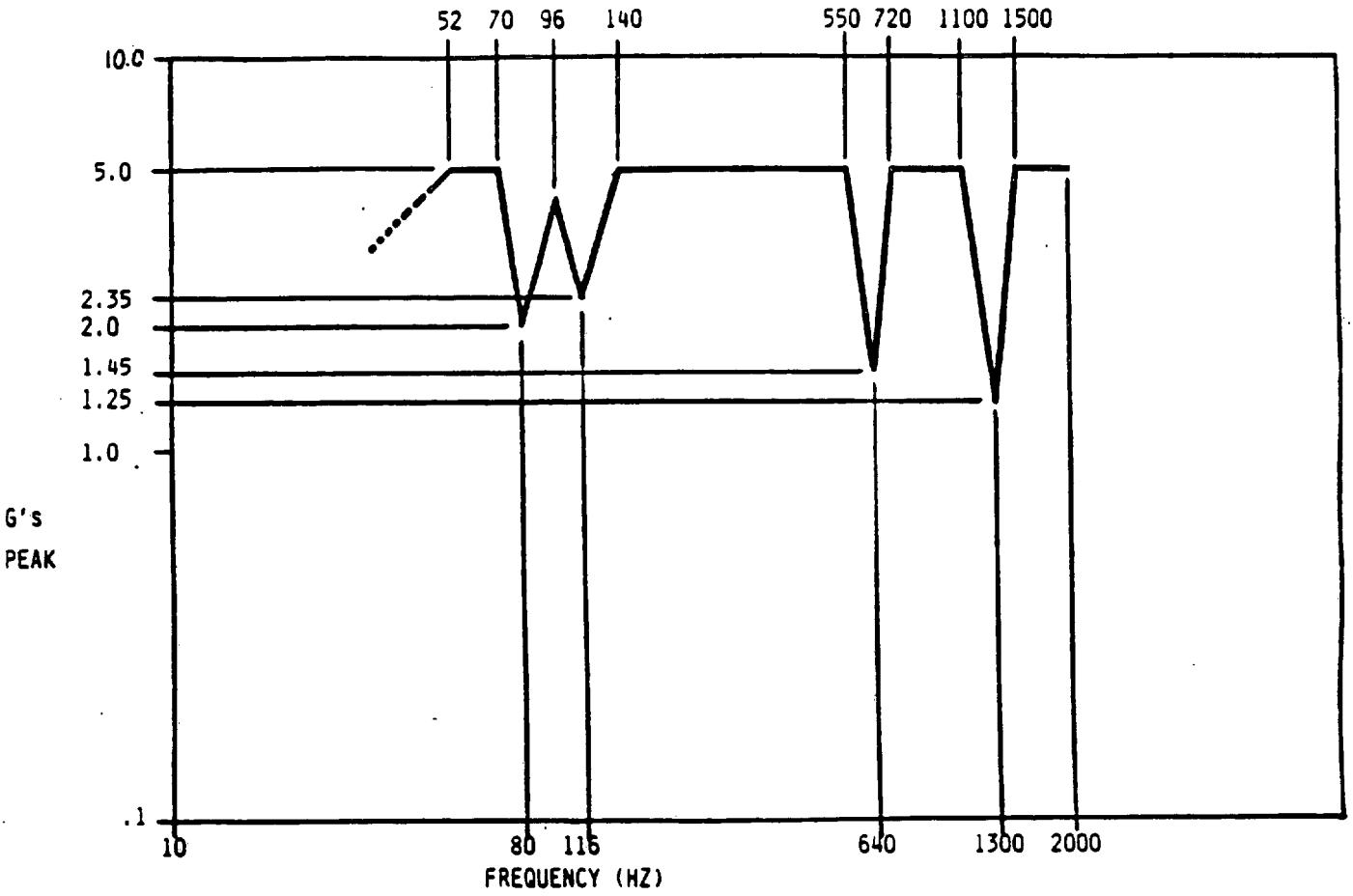


FIGURE 1. Sinusoidal Vibration - Vertical Plane (Functional)

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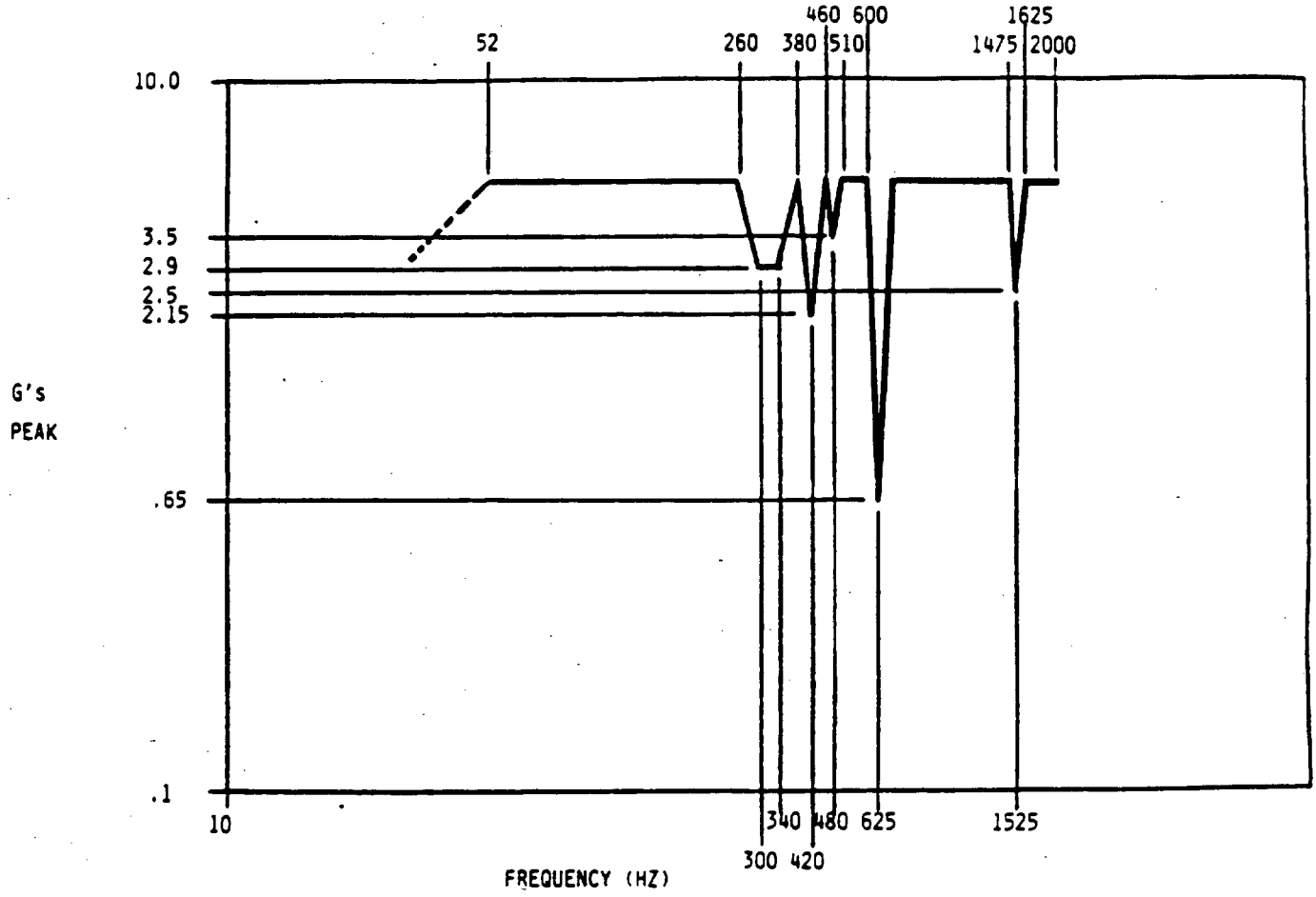


FIGURE 2 - Sinusoidal Vibration - Logitudinal Plane (Functional)

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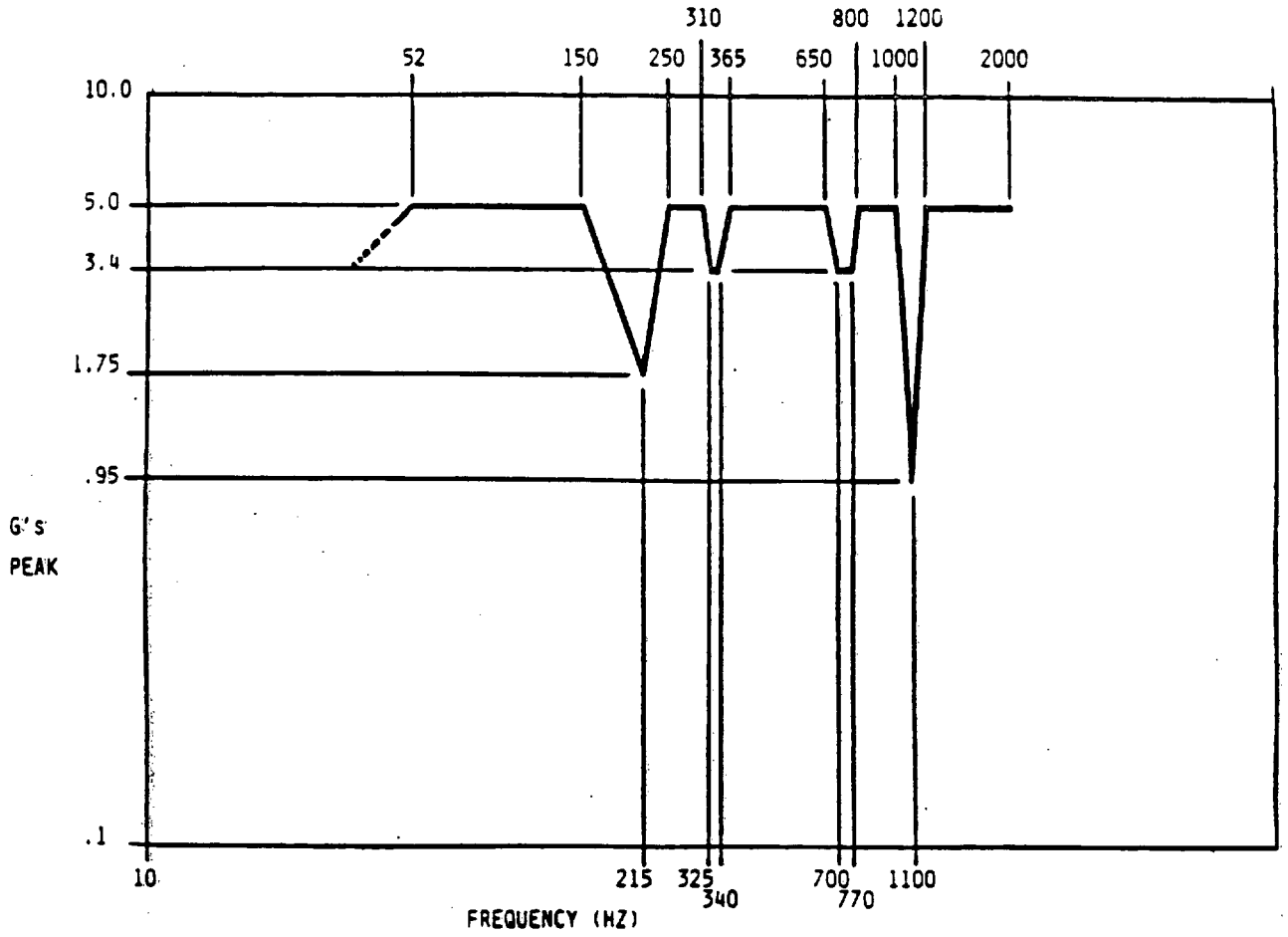


FIGURE 3. Sinusoidal Vibration - Lateral Plane (Functional)

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3.4.4.1 Service conditions. The receiver-transmitter shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for Class 2 equipmment, except as modified herein.

3.4.4.1.1 Sinusoidal vibration. The receiver-transmitter shall not require isolators or rear support and shall operate as required when subjected to sinusoidal vibration in accordance with test procedures of MIL-T-5422 to the requirements of Curve IIIB of MIL-E-5400 with the following exception. The sine sweep portion of the test above 52 Hz, which in Curve III B is 5 g's peak, shall be 5 g's peak except in the regions shown on Figures 1, 2, and 3 for the vertical, longitudinal, and lateral directions of vibration.

NOTE: If the number and nature of the break-cross overpoints in the sine sweep curve make the test unperformable on an automatic servo-controlled sine sweep generator, the test shall be performed with slow manual sweeping of the individual frequency bands so that approximately 15 minutes be required to sweep from 52 Hz to 2000 Hz.

3.4.4.1.2 Random vibration. The receiver-transmitter shall operate as required when subjected to the random vibration requirements specified in 4.4.5.

3.4.4.1.3 Rain protection of console mounted equipments. The portion of the receiver-transmitter which is normally exposed when the equipment is mounted in a console shall meet the rain requirements of MIL-STD-810, Method 506.2.

3.4.4.1.4 Thermal conditions. The receiver-transmitter shall operate as required under the thermal conditions specified by MIL-E-5400, Class 2, without the requirement for auxiliary cooling.

3.4.4.1.5 Temperature shock. The receiver-transmitter shall meet the temperature shock requirements of MIL-T-5422.

3.4.4.2 Warm-up time. The time required for the receiver-transmitter to warm up prior to operation shall be kept to a minimum and shall be not greater than 10 seconds at standard conditions. Under service conditions, the warm-up time shall be not greater than 20 seconds.

3.4.4.3 Electromagnetic interference (EMI). The receiver-transmitter shall comply with MIL-STD-461 for Class 1/A equipment except as modified in Tables III, IV, and herein. These requirements shall be met under all combinations of operating modes and frequencies of the receiver-transmitter.

3.4.4.3.1 Conducted and radiated emissions limits. The conducted and radiated emissions of the receiver-transmitter shall comply with the limits of the following requirements of MIL-STD-461:

a. CE01

b. CE02

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TABLE III. Exceptions to MIL-STD-461

CE03/04	Conducted Emissions, 20 KHz to 50 MHz Up to 6 dB above spec. Limit at frequencies between 2 MHz and 6 MHz for narrowband emissions. Transmit: Emissions up to 10 dB above spec limit of frequencies between 80 KHz and 210 KHz.
CE06	Conducted Emissions, Antenna, 10 KHz to 12.4 GHz Transmit: Compliance with paragraph 3.6.1.24.12, page 43, Table XIV transmitter harmonic limits. Receive: First injection frequencies and harmonics up to 10 dB above limit. Second injection harmonics up to 10 dB above limit.
RE02	Radiated Emissions, 14 KHz to 10 GHz Transmit: Transmit frequency for reference only. Harmonics of transmit frequency up to 15 dB above limit. Receive: No exceptions.
CS02	This requirement shall apply to all neutrals and grounds whether or not they are internally returned to chassis. In TR&G Receive Mode, the signal susceptibility requirement may be reduced by 10 dB.
CS04/08	Conducted Susceptibility, Receiver Spurious VHF Bands: Up to 15 dB below limit at UHF frequency associated with VHF channel frequency. Up to 20 dB below limit for frequencies outside of 30 to 400 MHz band. Up to 15 dB below spec. Limit for 1/2 frequency in low VHF-FM band (Fo = 60 MHz with Fx = 30 MHz).
RS03	Radiated Susceptibility, Electric Field Receive: Susceptibility limit reduced to 0.1 volts/meter at frequencies offset ± 29 MHz or ± 58 MHz from harmonics of the first and second injection frequencies. Transmit: No exceptions.
High Level RS03	High level RS03 shall be in accordance with the field levels and modulation as specified in Table IV.

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TABLE IV. High level RS03 signals

Frequency (MHz)	Modulation				Power Density (mw/cm^2) 5/	v/m (ref) 6/	Scan Rate (sec) 7/	Dwell Time (msec) 7/
220 8/	PAM1/	10 usec2/	PW3/	200 pps4/	10.6	200	4	500
220	PAM	200 usec	PW	200 pps	10.6	200	4	500
425	PAM	15 usec	PW	300 pps	10.6	200	10	200
900	PAM	2 usec	PW	1000 pps	10.6	200	5	50
900	PAM	125 usec	PW	300 pps	10.6	200	5	50
1250	PAM	10 usec	PW	3000 pps	10.6	200	5	50
1350	PAM	5 usec	PW	3000 pps	10.6	200	5	50
3000	PAM	3 usec	PW	1000 pps	10.6	200	5	50
3000	PAM	70 usec	PW	200 pps	10.6	200	5	50
3600	PAM	1 usec	PW	1000 pps	10.6	200	5	50
5600	PAM	1 usec	PW	600 pps	10.6	200	5	50
9100	PAM	0.5 usec	PW	1000 pps	10.6	200	5	50
9375	PAM	0.2 usec	PW	1000 pps	10.6	200	5	50
13900	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
15500	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
33200	PAM	0.2 usec	PW	2000 pps	10.6	200	0	CONT

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is defined as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Scan and Dwell refer to simulated antenna rotation. The signal shall be disabled except for the specified dwell. The signal shall be periodically enabled for the specified dwell time with the period equal to the specified scan time.
8/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations shown.

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c. CE03

d. CE04

e. CE06

f. RE01

g. RE02

3.4.4.3.2 Conducted and radiated susceptibility limits. The receiver-transmitter shall comply with the conducted and radiated susceptibility limits of the following requirements of MIL-STD-461:

a. CS01

b. CS02

c. CS03 (upper test frequency limit of 1.5 GHz)

d. CS04 (upper test frequency limit of 4 GHz)

e. CS05

f. CS06

g. CS08 (upper test frequency limit of 4 GHz)

h. RS01

i. RS02

j. RS03

The receiver-transmitter shall exhibit no susceptibility as defined in 3.4.4.3.3 when subjected to an RS03 test to the limits of MIL-STD-461. In addition, the EMC System shall exhibit no susceptibility in accordance with the criteria of 3.4.4.3.3b when subjected to signals at the levels of Table IV.

3.4.4.3.3 Susceptibility. The following criteria shall be applied to determine whether the receiver-transmitter is susceptible.

a. Any failure of the receiver-transmitter to comply with requirements of 3.6.1.13 and 3.6.1.14 shall be considered an indication of susceptibility.

b. Any repeatable non-commanded change in the modes of operation or preset frequencies of the receiver-transmitter shall be considered an indication of susceptibility. This shall apply whether the failure is transient (momentary) or non-transient (permanent) in nature. In addition the inability to change radio frequency or mode of operation while exposed to the susceptibility signal be considered an indication of susceptibility.

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3.4.4.3.4 Ground plane interference. All receiver-transmitter interface circuits shall be capable of specified performance when subjected to the following aircraft chassis noise between the interfacing WRA's.

a. Three volts rms from 320 Hz to 500 Hz (not to be greater than 150 ma rms applied current).

b. One volt rms from 500 Hz to 20 MHz (not to be greater than 150 ma rms applied current for frequencies between 500 Hz and 50 kHz. Not to be greater than 1 watt (CW) from a 50 ohms source for frequencies above 50 kHz.)

c. ± eight volt pulses, 100 usec wide at 100 pps (not to be greater than 15a peak current spikes).

In addition to the susceptibility specified in 3.4.4.3.3, the radio shall maintain intelligible communications during a "talk thru" test with another radio set using a low level signal (not to exceed 20u volts input to the receiver).

3.4.4.3.5 Conducted susceptibility test, interconnecting and signal leads. The unit shall exhibit no susceptibility, as specified in 3.4.4.3.3 when subjected to 100 mv rms 2 to 30 MHz coupled into its interconnecting and signal leads.

3.4.4.3.6 Grounding requirements. The grounding system between WRA's of the equipment shall be compatible and consistent with the following aircraft grounding scheme specified in 3.4.4.3.6.1 through 3.4.4.3.6.3. All WRA's shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, provide effective shielding for signal circuits, minimize EMI, and to protect personnel from electrical hazards.

3.4.4.3.6.1 Chassis grounds. A ground wire of minimum length connected internally to the WRA chassis shall be provided at a pin on each primary power connector. No circuit shall be allowed to utilize this wire as its primary return.

3.4.4.3.6.2 Shielding grounds. A separate connector pin shall be provided for each aircraft wire shield unless suitable grounding is provided in external cable backshells. The connector shield pins shall be grounded to the equipment chassis inside the WRA adjacent to the connector mounting and by the shortest means practical.

3.4.4.3.6.3 Component grounds. All externally exposed metal parts, shields, control shafts, switch handles, connectors, bushings, shall be grounded to the chassis.

3.4.4.3.7 Shielding gaskets. Shielding gaskets and similar devices for RF applications shall avoid the use of materials which represent severe dissimilar metal combinations with the interface.

3.4.5 Dimensions. The dimensions of the receiver-transmitter shall be as shown on Figures 4, 5, and 6.

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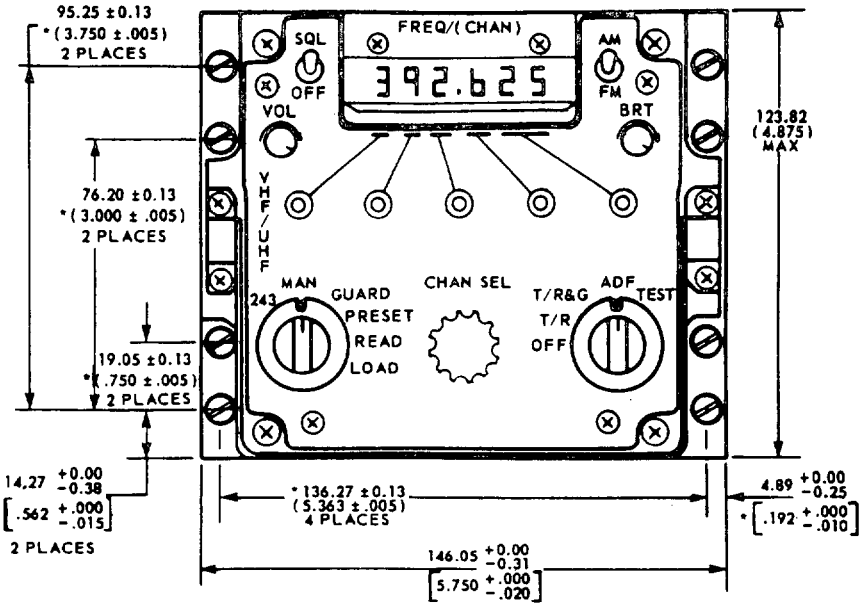


FIGURE 4. Front view, panel mount, receiver-transmitter

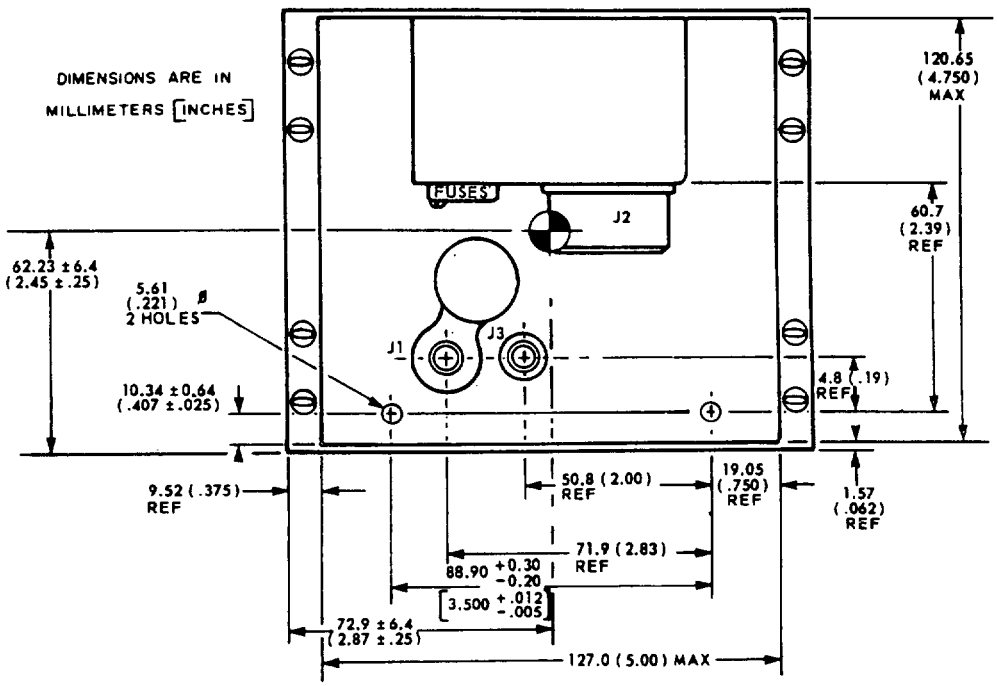


FIGURE 5. Rear view, panel mount, receiver-transmitter

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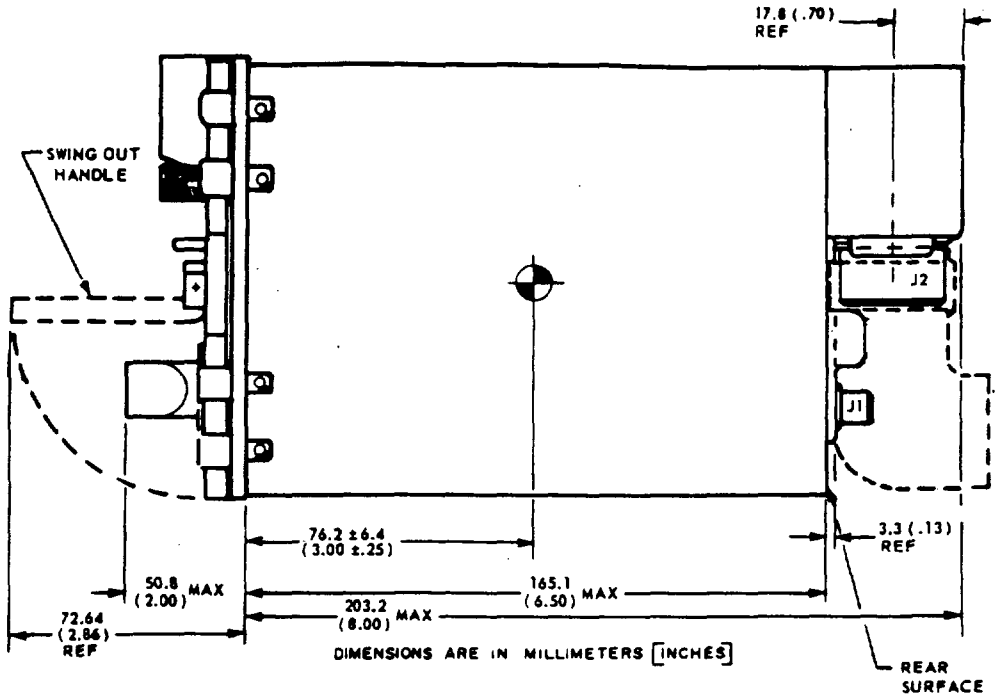


FIGURE 6. Side view, panel mount, receiver-transmitter

3.4.6 Weight. The weight of the receiver-transmitter shall be not greater than 10 pounds.

3.4.7 Color. The exterior color of the receiver-transmitter shall be lusterless black. Painted surfaces shall be in accordance with FED-STD-595, Color No. 37038 and anodized surfaces shall meet the requirements of MIL-A-8625 Type II, Class 2, Color Black.

3.4.8 Nameplates and identification marking. Nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.4.8.1 Decals, nameplates, and marking locations. All decals, nameplates, and marking shall be affixed to the receiver-transmitter as shown on Figures 4, 5 and 6.

3.4.8.2 Serial numbers. Serial numbers will be assigned by the procuring activity for the receiver-transmitter under contract or purchase order. The contractor shall serialize sequentially all receivers-transmitters which are delivered under the contract.

3.4.9 Input electrical power.

3.4.9.1 Operating power. The receiver-transmitter shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The voltage supplied shall be 28 volts DC. The power required shall be not greater than 20 watts in receive, 125 watts when transmitting at frequencies between 30 MHz and 174 MHz, or 150 watts when transmitting between 225 MHz and 400 MHz.

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3.4.9.2 Lighting power. Input power for lighting of any lighted panel shall be not greater than 5.0 watts at 5 or 28 volts.

3.4.9.3 Degraded performance. Degraded performance will be permitted for voltage transients with durations of 75 microseconds to 0.5 seconds during normal electrical system operation. Operation shall return to normal within five seconds with no resulting damage to the receiver-transmitter.

3.4.9.4 Emergency voltage. During operation in the emergency steady state electric power condition as specified in MIL-STD-704, the receiver-transmitter shall remain safe and frequency readouts and control functions shall be maintained. The emergency power source will impose an interval of zero voltage at time of loss of primary power for a maximum period of 30 seconds. The receiver-transmitter is not required to perform during this period of power loss but shall return to performance as specified above after return to emergency power conditions in accordance with MIL-STD-704. Performance characteristics at emergency limit voltage shall be:

a.	RF power output:	2 watts minimum
b.	Receiver sensitivity	-97 dBm AM and -104 dBm FM maximum for 10 dB signal plus noise-to-noise ratio
c.	Audio output power:	6 dB maximum below standard conditions

3.4.9.5 Undervoltage protection. The receiver-transmitter shall not be damaged by voltages below the minimums specified by MIL-STD-704, and shall be capable of resuming normal operation when the voltage returns within normal limits.

3.4.9.6 Reverse polarity protection. The receiver-transmitter shall be protected such that it will not be damaged by application of reverse polarity voltages.

3.4.10 Manufacturing requirements. The receiver-transmitter shall be manufactured in accordance with manufacturing processes, workmanship, and quality assurance provisions of MIL-STD-454, MIL-STD-275, MIL-P-55110, WS-6536, NAVMAT P-4855.1, and NAVMAT P-9492.

3.5 Performance. Unless otherwise specified herein, values set forth to establish specified performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified herein.

3.5.1 Operation. The receiver-transmitter shall provide the functions and capabilities specified in 3.1 for operation in military aircraft and helicopters. Additionally the receiver-transmitter shall be compatible with associated equipment listed in 6.5 to provide operations specified in 3.6.

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3.6 Detail requirements.

3.6.1 Receiver-transmitter, panel mount. The panel mount receiver-transmitter shall contain provisions, including control functions, for transmit and receive communication. By one wire "take control" switching, the receiver-transmitter shall be operatable by slow speed serial data bus from a remote control unit as specified in MIL-C-85666, such as the C-10319A/ARC. The slow speed serial data characteristics shall be in accordance with MIL-C-85666.

3.6.1.1 Function. The receiver-transmitter shall provide simplex two-way communication among tactical platforms in the frequency bands as specified in a through n. Channel spacing in all bands shall be 25 KHz.

a. Low VHF (FM). 30.000 MHz through 87.975 MHz, frequency modulation, normal voice, secure voice, squelch tone transmission, and slow frequency hop.

b. VHF (AM). 108.000 MHz through 155.975 MHz, amplitude modulation, normal voice, secure voice, automatic direction finding (ADF), 1020 Hz tone transmission and slow frequency hop. Transmit shall be inhibited from 108.000 through 117.975 MHz.

c. VHF (FM). 156.000 MHz through 173.975 MHz, frequency modulation, normal voice, secure voice, automatic direction finding (ADF), 1020 Hz tone transmission, and slow frequency hop.

d. UHF (AM). 225.000 MHz through 399.975 MHz, amplitude modulation, normal voice, secure voice, automatic direction finding (ADF), 1020 Hz tone transmission and slow frequency hop.

e. UHF (FM). 225.000 MHz through 399.975 MHz, frequency modulation, normal voice, secure voice, automatic direction finding (ADF), 1020 Hz tone transmission, and slow frequency hop.

f. Continuous monitoring of the guard frequency in the selected band. The guard frequency for each band is listed in Table V.

TABLE V. Guard frequency for each band.

Range	Guard Channel
30-87.975 MHz	40.5 MHz-FM
108-155.975 MHz	121.5 MHz-AM
156-173.975 MHz	156.8 MHz-FM
225-399.975 MHz	243.0 MHz-AM

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g. Compatibility with TSEC/KY-28 and TSEC/KY-58 secure voice equipment.

h. Automatic relaying whereby interconnection of two radio sets results in automatic transmission on one radio set of the signals being received by the other radio set.

i. Slow serial data interface for display of the receiver-transmitter operating frequency or preset channel number on remote indicators.

j. Guard channel precedence operation whereby an external control automatically turns the radio on and tunes both the transmitter and main receiver to the 243.000 MHz (AM) guard frequency.

k. Direct replacement of the AN/ARC-159(V) radio set in regard to form factor and interface functions such that the radio sets will maintain the capabilities of the AN/ARC-159(V) but will operate in all the frequency bands from 30 to 400 MHz.

l. Operation with equipment listed in 6.5 or equal.

m. Transmission of signals from a reference signal generator of CASS or DICASS on an assigned channel.

n. Compatibility with HAVE QUICK when controlled by C-11128/ARC, C-11129/ARC and C-11130/ARC in accordance with MIL-C-85806.

3.6.1.2 Controls. Control of the functions specified herein shall be provided by the front panel facilities or the slow serial interface along with parallel control lines as specified in the 3.6.1.18.

3.6.1.3 Operation modes.

a. OFF. In this mode, the internal DC power shall be electronically switched/removed from all but the on/off circuitry of the radio set.

b. Main receiver-transmitter mode (T/R). In this mode, the main receiver shall be turned on and the guard receivers shall be in the OFF condition. The transmitter shall be in the ready (key up) condition.

c. Main receiver-transmitter guard mode (T/R+G). In this mode, both main and guard receivers shall be turned on and the operator shall be able to hear the audio output of both receivers simultaneously. The guard receiver selected shall be that one which is in the frequency band of the main receiver frequency selected. Unless otherwise specified in the contract, the radio set shall have the same transmit-receiver capabilities as specified in the main receiver-transmitter mode (see 6.2.1).

d. ADF mode. In this mode, the main receiver-transmitter and guard receiver shall be operating as specified in 3.6.1.3c above, plus the radio set shall be capable of operation with the associated automatic direction finder equipment.

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e. Built-in-test. In this mode a performance check shall be run to verify radio set operation. The GO/NO-GO results of the BIT shall be returned to the remote control and displayed on the front panel.

3.6.1.4 Frequency select modes. The frequency select modes shall allow the operator to select emergency, guard, manual, or preset modes of frequency selection. In addition, provisions shall be made for the preset loading and preset read functions.

a. Emergency mode (243). In this mode, the receiver-transmitter shall be turned on and tune the transmitter and the main receiver to the 243.000 MHz (AM) guard frequency. When operating in this mode, all other controls shall be inoperative. This mode shall have precedence over all other modes.

b. Manual mode. This mode shall allow the operator to rapidly and easily select any one of the possible operating frequencies. When operating in the manual mode, it shall be impossible to change or affect the operating frequency of the radio set by means of the preset frequency select control. The transmitter and receivers shall be disabled while changing frequencies.

c. Guard mode. In this mode, the receiver-transmitter shall automatically tune the main receiver and transmitter to the guard channel of the selected frequency band and shall read out the appropriate frequency on the display specified. The selected frequency band shall be considered to be the frequency band that incorporates the frequency or channel last selected on either the preset (if the radio was last used in the preset mode) or the manual frequency controls (if the radio was last used in the manual mode). When operating in this mode, the preset frequency selector control as well as the manual frequency selector shall be inoperative.

d. Preset mode. This mode shall allow the operator to rapidly and easily select by only one control 28 preset operating channels by use of the control as specified in 3.6.1.1.10. The channel number (1 through 28) and the frequency of any operating channel selected shall be shown on the display as specified in 3.6.1.11. When operating in this mode, it shall be impossible to change the operating frequency of the radio set by means of the manual frequency selector controls. The transmitter and receivers shall be disabled while changing frequencies.

e. Preset read. This mode shall display the preset channel frequency to which the receiver-transmitter is tuned in lieu of the preset channel number. The frequency will be shown on the display as specified in 3.6.1.11. It shall be possible to select and display (but not load or change) any proper frequency in this mode. Changing and loading of the preset frequency shall be accomplished by using the preset load mode only.

f. Preset load. The mode shall be provided to transfer the selected frequency to the desired channel in the preset memory. The selected frequency for loading may be either the frequency last selected or that displayed in either the manual or panel read mode. The position shall be designed in such a way as to prevent unintentional loading.

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3.6.1.5 Improper frequencies. Under no conditions shall the receiver-transmitter tune to improper frequencies. These frequencies are: below 30.000 MHz, between 88.000 through 107.975 MHz, between 174.000 through 224.975 MHz and above 399.975 MHz. In the frequency range from 108.000 MHz through 117.975 MHz where the receive mode is proper, the equipment shall not permit transmission.

3.6.1.6 Audio volume control. The audio volume control shall allow the operator to manually adjust the audio power output level. The control shall provide maximum output in the full clockwise position with attenuation increasing at least 48 dB when rotating in the counterclockwise direction to approximately 95 percent of the rotation. In the full counterclockwise position (100 percent), no audio shall be present. The attenuation change, in dB, shall be essentially linear with degree of rotation.

3.6.1.7 Squelch control. The squelch control shall allow the operator to disable the squelch circuit on the main receiver only. In the off condition the squelch circuit shall be disabled to allow receiving of main receiver signals regardless of carrier to noise characteristics. A pin on the rear connector (J2) shall be provided which will disable the main squelch circuit when the pin is grounded. Grounding the pin shall not affect the guard receiver squelch.

3.6.1.8 Light dimmer control. A light dimmer control shall be provided on the front panel to adjust the illumination intensity of the display described in 3.6.1.11.

3.6.1.9 Manual frequency select controls. Control shall be provided on the front panel to manually select any of the radio frequencies.

3.6.1.10 Preset frequency select control. One selector control shall be provided on the front panel for selection of any one of the 28 available preset channels.

3.6.1.11 Frequency or channel display. A six digit frequency display shall be provided on the front of control to display the operating radio frequency. A decimal shall also be lighted to identify the MHz position of three significant digits (for preset channel number 1 through 28) shall be lighted with the remaining digits unlighted. The display shall be arranged horizontally in a centrally located position on the front panel of the radio set. The displays shall be such that the digits can be easily read from a distance of 2.0 feet at any angle up to at least 30 degrees from the display centerline perpendicular to the surface of the front panel. Provisions shall be made for easy removal and replacement of the display digits without loosening or removing the equipment front panel or control knobs. Additionally, the control shall provide serial data output to operate equivalent displays of a panel mount radio set remote indicators.

3.6.1.12 AM or FM select. In all bands except the UHF band the AM or FM modes shall be automatic to provide: FM in the 30 to 88 MHz band, AM in the 108 to 156 MHz band, and FM in the 156 to 174 MHz band. For the UHF band the AM or FM select switch shall be easily accessible to choose the AM or FM mode. This switch shall be configured such that unintentional switching is precluded.

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3.6.1.13 Take control function. A pin shall be provided in the rear connector (J2) which shall, when grounded, disable the front panel radio operation controls and permit control from a C-10319A/ARC, or equal unit, via the slow serial data bus.

3.6.1.14 Memory device. The memory device shall be a nonvolatile type such that power off, power line interruptions or any combination of environmental conditions will not cause a loss or change of operating frequency and mode.

3.6.1.15 Frequency chart. A frequency chart shall be provided to allow maintenance personnel or the operator to record the channel frequency for each and all of the preset channels. The chart shall be preprinted showing sequentially the preset channel numbers and providing space to record the associated frequency. The surface of the chart shall be durable such that a pencil entry may be clearly recorded and erased for a minimum of 1000 times. The chart shall be approximately 3 inches by 5 inches. A separate chart shall be packaged with each radio set, receiver-transmitter and radio set control and shall have an adhesive backing for mounting elsewhere in the cockpit.

3.6.1.16 Main and guard receiver controls. Screwdriver adjustable squelch level controls shall be accessible on the main and guard receivers without opening the receiver-transmitter. A screwdriver adjustable audio output control shall be accessible on the main receiver without opening the receiver-transmitter (see 3.6.1.23.3).

3.6.1.17 Duty cycle. The receiver-transmitter shall have at least a one minute-five minute receive duty cycle. For extended periods of transmit operation (in excess of duty cycle), thermal sensing periods of the transmitter shall automatically limit the power dissipation of the transmitter to maintain safe operating temperatures and prevent damage to the unit. The radio set shall contain a thermal circuit for protection against the overheating of the transmitter power amplifiers. If, by inadvertently leaving the receiver-transmitter in the transmit condition or by some other means, the transmitter power amplifiers become overheated, the thermal protective circuit shall so reduce the dissipated power of the power amplifiers so that no damage will occur to any component part and the radio-transmitter will operate as required again when the temperature of the power amplifier is returned to normal. The available RF power output under extended time transmission shall be two watts minimum.

3.6.1.18 Electrical connections. Connections to external circuits shall be in accordance with Table VI.

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TABLE VI. Connections to external circuits.

Reference Designation	Receptacle Type	Function
J1	TNC M39012/31-001	Comm. Antenna
J2	Bendix P/N 21-529017-200	Power/Control
J3	TPS-Bendix P/N 7034-7	ADF

The receiver-transmitter shall communicate with its associated controls as specified in 3.6.1.8.1 and 3.6.1.8.2.

3.6.1.18.1 Parallel control lines . The receiver-transmitter shall interface with other equipment in the system using the parallel lines of J2 as specified in Table VII. The parameters listed are nominal. For specified limits, see the applicable paragraph.

3.6.1.18.2 Serial data bus. The serial data bus shall be used to communicate between the remote controls, indicators, and host receiver-transmitter. The serial data bus shall consist of a shielded twisted pair for the data bus and a shielded twisted pair for the clock bus. The characteristics of the lines shall be as specified in Table VII (pins J2-18, 19, 26 and 27). Further definition of the serial interface is provided in MIL-C-85666.

CAUTIONARY NOTE
ARC-182 SERIAL CONTROL INTERFACE SPECIFICATION

All devices connected to the serial data bus should read and decode all data information. Further, the device should decode command word information to compare data sent versus data responses, to insure that no inconsistencies have occurred, either as a result of false clock pulses, bit transitions or external interference.

3.6.1.19 Transmit frequency accuracy. The transmitted RF carrier frequency shall be within ±1.5 part per million (±1.5 Hz per MHz) of the selected operating frequency at the time of delivery under service conditions. From +71°C to +95°C ambient temperature, the carrier frequency shall be within ± 10 parts per million (± 10 Hz per MHz) of the selected operating frequency.

3.6.1.19.1 Aging. The change in transmitter radio frequency due to aging shall be less than or equal to 0.5 part per million per year with an adjustment capability for eight years.

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TABLE VII. RT-1324A/ARC, RT-1327A/ARC, RT-1407/ARC power/control connector (J2) interface characteristics 1/

Pin No.	I/O	Function	Characteristics
1	I	Front Panel	2/ 5 volts AC or DC 5 watts maximum.
2,3	I	Input Power	+28 VDC in accordance with MIL-STD-704 150 watts maximum in transmit 20 watts maximum in receive.
4	I/O	System On/Off	0 to 7 VDC turns receiver-transmitter on, open circuit turns receiver-transmitter off. 25 mA current sink capability required of external switch if used. Open circuit DC resistance shall be > 100K ohms.
5	I	Squelch Disable	Ground or 0 VDC ± 5 VDC disables main receiver squelch. Open circuit enables main receiver squelch.
6	I	Tone Transmit	Ground or 0 VDC ± 5 VDC keys transmitter with 1000 Hz tone modulation. Open circuit for not tone transmit.
7	I	Front Panel Lighting	2/ 28 volts AC or DC. 5 watts maximum.
8,9	I	Ground	Ground return for input power and signal circuits.
10		Spare	
11	I	External Sync	Use with HAVE QUICK control to strobe data into receiver-transmitter ground or 0 to + 0.5 VDC = no data transfer. Open=data transfer. Pulse duration > 20 microseconds. Function is used only in conjunction with pin 41. When HAVE QUICK control is not connected, pin 41 must be HI.
12	-	Spare	
13	I	Guard Precedence	Ground or 0 ± 5 VDC turns on receiver-transmitter to 243 MHz, open for normal operation. Open circuit impedance shall be > 100K ohms.

1/ This Table indicates nominal characteristics for specification limits see appropriate paragraphs, herein.

2/ Use either pin 1 or pin 7 (not both).

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TABLE VII. RT-1324A/ARC, RT-1327A/ARC, RT-1407/ARC power/control connector (J2) interface characteristics - Continued

Pin No.	I/O	Function	Characteristics
14	-	Panel Lamp Power Common	
15	I	Receiver Sensitivity	Ground or 0 VDC + 1 VDC reduces receiver sensitivity by 10 dB in frequency range of 108 to 174 MHz. Open circuit for full sensitivity.
16	O	UHF Band Indicator	Open collector transistor output capable of sinking up to 100 mA. At Eo ≤ 1 VDC for UHF mode. Open circuit in all VHF bands.
17	I	X-Mode Audio Select	Ground or 0 VDC + 5 VDC for X-mode audio select. Open circuit for normal operation. 40 mA current sink capability required of external switch contact.
18	I/O	Clock (HI)	Active only when control information is being sent, "1"=4.8 VDC + 0.5 VDC. "0"=0 VDC + 0.5 VDC. 2.7K maximum DC ohm source impedance, 22K load minimum impedance. Line is at "0" when not active.
19	I/O	Data (HI)	Same as pin 18, except line is at "1" when not active.
20	O	ADF Indicator	Same as pin 16 except ground (≤1 VDC) is provided when in ADF mode.
21	-	Spare	
22	-	Spare	
23	O	Low VHF-FM Band	Same as pin 16 except ground (≤1 VDC) is indicator provided for low VHF-FM band operation.

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TABLE VII. RT-1324A/ARC, RT-1327A/ARC, RT-1407/ARC power/control connector (J2) interface characteristics - Continued

Pin No.	I/O	Function	Characteristics
24	-	Spare	
25	I	Take Control-Normal	Open circuit=front panel controls effective. Ground=receiver-transmitter under remote control.
26	I/O	Clock (LO)	Same as pin 18 except line is at "1" when not active.
27	I/O	Data (LO)	Same as pin 18 except line is at "1" when not active.
28	O	AM Mode Indicator	Same as pin 16 except ground (≤1 VDC) is provided when receiver-transmitter is in AM mode.
29	O	Wideband Receive Audio Out	0.25 VRMS to 8.3 VRMS into 20K ohms +2, -4 dB from 30 Hz to 25 KHz.
30	I	Transmit Blanking	Ground or 0 VDC ± 1 VDC inhibits transmitter. Open enables normal transmitter controls.
31	O	ADF Audio Out	4 VPP minimum into 20K ohms; frequency response ± 2 dB from 50 Hz to 10 KHz.
32	-	Spare	
33	-	Spare	
34	I	Bandwidth Control	Ground or 0 VDC ± 5 VDC=wideband open or > 10 VDC=narrowband 5 mA current sink capability required of external switch.
35	-	Spare	
36	O	Voice Output (150)	High side of 150 ohm balanced output. Pin 44 is common side.
37	O	Voice Output (600)	High side of 600 ohm balanced output. Pin 44 is common side.
38	O	Guard Low Level Output	0.15 VRMS to 0.45 VRMS into 600 ohm load, unbalanced.

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TABLE VII. RT-1324A/ARC, RT-1327A/ARC, RT-1407/ARC power/control connector (J2) interface characteristics - Continued

Pin No.	I/O	Function	Characteristics
39	O	Retransmit Control	Same as pin 16 except ground (<1 VDC) is (key) provided when received signal strength is sufficient to break squelch.
40	I	Retransmit Audio In	6.5 VRMS into 1000 ohms provide 90 percent AM or ± 5.6 KHz FM deviation.
41	I	Take Control-Special	Used for HAVE QUICK mode of operation. Ground or 0 VDC ± 1 VDC=HAVE QUICK open=normal mode.
42	O	Retransmit Audio Out	6.5 VRMS ± 1 VRMS into 1000 ohms for 30 percent AM or ± 2.4 KHz FM deviation with 250 mW output. Output impedance = 60 ohms maximum.
43	I	Push to Talk (PTT)	Ground or 0 VDC ± 5 VDC-transmitter keyed open or > 10 VDC-receive mode.
44	O	Voice Output (Common)	Common side of balanced voice output.
45	O	Station Approach	Unbalanced DC meter drive current of 150 microamperes at input signal strength of -30 dBm; load impedance= 1000 ohms. 0 microamperes at approx. -100 dBm.
46	O	GD if AGC	Test point - positive going voltage between 1 VDC and 7 VDC for increasing signal strength.
47	I	Mic Audio In (LO)	Common side of microphone input. Grounded in receiver-transmitter.
48	I	Wideband Transmit Audio In	12 VPP ± 1V into 200 ohms produces 90 percent AM or ± 5.6 KHz FM deviation.
49	I	Maritime Offset	Ground or 0 VDC ± 5 VDC causes RT to operate in half-duplex mode with 4.6 MHz offset on selected frequencies in the maritime band. Open or 10 VDC= normal simplex operation.
50	-	Spare	

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TABLE VII. RT-1324A/ARC, RT-1327A/ARC, RT-1407/ARC power/control connector (J2) interface characteristics - Continued

Pin No.	I/O	Function	Characteristics
51	I	X-Mode Receive Audio In	Deciphered audio from secure voice equipment. 6.5 VRMS into 110 ohms required for 250 mW output.
52	O	Unattenuated Audio Out	Same as pin 42 except is disconnected from receiver-transmitter audio circuits when pin 17 is grounded.
53	I	Mic Audio In (HI)	Adjustable 0.25 VRMS to 1.5 VRMS for 90 percent AM or ± 5.6 KHz FM deviation. Input impedance=175 ohm ± 25 percent normally adjusted for 0.5 VRMS input.
54	-	Spare	
55	I	Ground	Chassis ground in receiver-transmitter.

3.6.1.19.2 Frequency accuracy adjustment. The transmitter radio frequency shall be adjustable to within ± 1 part in 10^8.

3.6.1.20 Channel changing time. The time from the instant any channel is selected (after warm-up and including band switching time) frequency controls to the time that the radio set is ready on the selected frequency for either transmission or reception shall be not greater than 65 milliseconds.

3.6.1.20.1 Transceiver frequency transition time. The transceiver frequency settling time as controlled by an external applique shall be not greater than four milliseconds for channel changes within any band.

3.6.1.21 Spontaneous channel change. When operating there shall be no spontaneous change of channels when the equipment is subject to any combination of environmental conditions or electrical input. Turning the radio off shall not cause a change in frequency when the radio is turned back on.

3.6.1.22 Antenna circuits. Two antenna ports shall be provided. The main receiver, transmitter, and guard receivers shall normally all operate with the same antenna through connector J1. The receivers shall be so isolated during transmission that no damage or degradation in performance shall result from using a common antenna. No damage shall occur to any part of the radio set when the transmitter is operated with the RF output shorted, open-circuited or terminated in any complex impedance. An RF connector shall be provided for automatic direction finding. When the radio is in the ADF mode and tuned to the lower (30-88 MHz) VHF (FM) band, the main receiver shall be connected to antenna port J3. When the radio is in the ADF mode and tuned

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to any band other than the lower VHF (FM) band, the main receiver shall be connected to antenna port J3. The transmitter shall be connected in all cases to J1. The guard receivers shall remain operational in the ADF mode.

3.6.1.22.1 External antenna switch. The receiver-transmitter shall provide a ground or open on a separate pin of connector J2 for control of an external antenna switch SA-521/A (MIL-S-25879) or an appropriate equal. A ground shall be provided on this pin only when in the ADF mode and when not operating in the 30-88 MHz band. This pin shall be open circuited in all other conditions. (See 3.6.1.27.b)

3.6.1.23 Receiver characteristics. Unless otherwise specified in the contract, the following receiver characteristics shall apply to both the main and the guard receiver (see 6.2.1). All signal levels (dBm) are at the antenna input terminals and assume a receiver impedance of 50 ohms.

3.6.1.23.1 Sensitivity. A (S + N)/N of not less than 10 dB shall be produced at the normal voice output of the receiver for the signal levels specified in Tables VIII and IX. The test signal shall be frequency modulated by 1 KHz at ± 2.4 kHz deviation for the FM bands and amplitude modulated 30 percent at 1 KHz for the AM bands. A pin on the rear connector shall be provided which shall reduce the sensitivity of the main and guard receiver in the 108 MHz to 174 MHz range by 10 dB when the pin is grounded.

3.6.1.23.2 Selectivity. The main receiver shall incorporate a narrowband and a wideband filter such that the receiver is capable of working with transmitters that have ± 10 KHz stability. Design and location of the filters shall be such that they can be replaceable at the intermediate maintenance level with filters with different bandwidth characteristics. Bandwidths are as follows:

a. Narrowband for normal voice and baseband secure.
Bandwidth: ± 17.0 KHz min. at 6 dB points
referenced to center frequency.
+ 35 KHz max. at 60 dB points
referenced to center frequency.

b. Wideband for diphase secure. Bandwidth:
± 34.5 KHz min. at 6 dB points
referenced to center frequency;
+ 85 KHz max. at 60 dB points
referenced to center frequency.

c. Guard receiver bandwidth:
± 14.0 KHz min. at 6 dB points
referenced to center frequency.
+ 40.0 KHz max. at 60 dB points
referenced to center frequency.

Pass band ripple shall be not greater than 3 dB over 80 percent of the specified 6 dB bandwidth.

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TABLE VIII. Main receiver sensitivity

Frequency	Mode	Signal Level	
		Standard Conditions	Service Conditions
30-87.975 MHz	FM	-112 dBm	-110 dBm
108-155.975 MHz	AM	-103 dBm	-101 dBm
		-93 dBm 1/	-91 dBm 1/
156-173.975 MHz	FM	-110 dBm	-108 dBm
		-100 dBm 1/	-98 dBm 1/
225-399.975 MHz	AM	-103 dBm	-101 dBm
225-399.975 MHz	FM	-110 dBm	-108 dBm

1/ Receiver sensitivity control pin grounded

TABLE IX. Guard receiver sensitivity

Frequency	Mode	Signal Level	
		Standard Conditions	Service Conditions
40.5 MHz	FM	-112 dBm	-110 dBm
121.5 MHz	AM	-103 dBm	-101 dBm
		-93 dBm 1/	-91 dBm 1/
156.8 MHz	FM	-110 dBm	-108 dBm
		-100 dBm 1/	-98 dBm 1/
243.0 MHz	AM	-103 dBm	-101 dBm

1/ Receiver sensitivity control pin grounded

3.6.1.23.2.1 Adjacent channel rejection. The main receiver shall be capable of 60 dB rejection of normal and FM secure voice signals on channels at least 50 KHz away with the specified narrow band IF selectivity. (See 3.6.1.23.2.)

3.6.1.23.3 Squelch. The main and guard receivers shall have squelch circuits that are compatible with CLIMAX and AN/PRC-90 swept tone type emergency beacons and that operate in the normal voice circuits for both AM and FM to allow a receiver output only when the received signal plus noise-to-noise ratio is at or above 5 dB. An adjustment shall be provided to select the signal plus noise-to-noise level of squelch operation over the range of 5 dB to 15 dB. The squelch circuits shall prohibit receiver output when the received signal plus noise-to-noise ratio is below the adjusted threshold level.

3.6.1.23.3.1 Squelch attack or release times. Squelch attack and release times shall be not greater than 150 milliseconds for a signal level of -70 dBm.

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3.6.1.23.3.2 Squelch overlap. Squelch overlap is specified as the difference of the RF voltage required to unsquelch the main and guard receiver outputs to the voltage at which the output is again squelched. This difference shall be not less than 1.5 dB and in no case shall a change of more than 12 dB be required.

3.6.1.23.4 Main and guard receiver audio outputs. Audio output tests specified in 3.6.1.23.4.2 shall be conducted with maximum audio output (250 mW across 600 ohms, see 3.6.1.23.4.1) unless otherwise required.

3.6.1.23.4.1 Normal audio output. The main and guard receivers shall have a common output. The audio outputs of the main and guard receiver shall be in phase. The common output shall be available at the unattenuated audio output as specified in 3.6.1.23.4.7 and also at the 150/600 balanced output. The audio output at the balanced output shall be 250 mW minimum with an input signal of -53 dBm with 30 percent AM or ± 2.4 KHz FM deviation. Volume control of the audio output is adjustable as specified in 3.6.1.5. Provision shall be made to set the audio output at any level between 100 mW and 400 mW using a screwdriver adjustment on the receiver module. Access to the adjustment shall not require disassembly of the receiver-transmitter. The adjustment is provided to accommodate variation in system requirements. The standard output as delivered from the factory shall be 250 mW minimum for 30 percent AM or ± 2.4 KHz FM deviation. Under service conditions the audio output shall be 200 mW minimum. From +71°C to +95°C ambient temperature, the audio output shall be 150 mW minimum.

3.6.1.23.4.2 Audio gain or level control. The automatic gain control (AGC) or automatic level control (ALC) shall maintain the main and guard receiver audio output level to within ± 3 dB over the following RF input ranges when referenced from -53 dB.

a. AM: -103 to +7 dBm

b. FM: -112 to +7 dBm

In addition, for a constant -53 dBm input the audio level change shall be not greater than 2 dB when changing from band to band or when changing between main and guard receivers.

3.6.1.23.4.3 Audio output protection. The receivers shall operate as specified after the audio outputs have been subjected to both open and short circuit conditions.

3.6.1.23.4.4 Bandwidth. The normal audio output frequency response shall be ± 3dB relative to 1 KHz reference for signals from 400 Hz to 3.0 KHz. Frequencies 150 Hz and below and frequencies 4.0 KHz and above shall be attenuated not less than 30 dB.

3.6.1.23.4.5 Normal audio output impedances. The normal audio output impedance shall be suitable for operation into both 150 and 600 ohm loads. This output shall be floating with respect to chassis ground. External grounding of one of the output leads shall have no adverse effects on receiver performance. Only one output shall be loaded for normal operation.

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3.6.1.23.4.6 Normal audio output distortion. Total harmonic distortion at the normal audio output when terminated by either 150 or 600 ohms shall be not greater than six percent over the RF input range with 30 percent modulation or ± 2.4 KHz deviation when the audio level is adjusted to maximum levels required. Under service conditions the distortion shall be not greater than 10 percent.

3.6.1.23.4.7 Unattenuated audio output. In addition to the normal audio output loaded into 150 or 600 ohms, an unattenuated, unbalanced audio output shall be provided on a separate pin on the external connector. With the 150 or 600 ohm output properly terminated, the unattenuated output level shall be 6.5 ± 1.0 volts rms when loaded with 1000 ± 10 percent ohms for an RF input of -53 dBm modulated 30 percent AM or deviated ± 2.4 KHz FM at 1 KHz. Under service conditions, the audio output level shall be no less than 5.0 VRMS. The frequency response shall be as specified in 3.6.1.23.4.4. Distortion shall be not greater than six percent at 1000 Hz with 30 percent modulation AM or ± 2.4 KHz deviation FM. Under service conditions the distortion shall be not greater than 10 percent. Source impedance of the output shall be not greater than 60 ohms. In systems where the audio output has been reduced (see 3.6.1.23.4.1), the unattenuated audio output will be reduced proportionally. For example: 100 mW audio output corresponds to 4.0 VRMS versus 6.5 VRMS at 250 mW.

3.6.1.23.4.8 X-mode receive audio. Provision shall be made to return deciphered audio from secure voice equipment through the audio interface of the receiver-transmitter. When the x-mode audio select line (J2-17) is grounded, the internal audio amplifiers shall be disconnected from the unattenuated audio output and the normal voice output, and replaced with deciphered audio from the secure voice equipment. An audio input of 6.5 VRMS into 110 ohms is required to provide the output levels specified in 3.6.1.23.4.1 and 3.6.1.23.4.7.

3.6.1.23.5 AM internal noise level. The signal plus noise-to-noise (modulated to unmodulated) ratio of the main and guard receivers shall be not less than 30 dB when a -53 dBm signal modulated 30 percent at 1,000 Hz is applied.

3.6.1.23.6 FM noise quieting. An RF signal of -70 dBm deviated ± 8.0 KHz by 1 KHz shall produce a signal plus noise-to-noise ratio of not less than 30 dB for both main and guard receivers.

3.6.1.23.7 Noise peak limiter. The detector of the main and guard receivers (following the intermediate frequency amplifier) shall be provided with an instantaneous noise peak limiter. The noise limit action shall effectively limit the audio output voltage of modulating surges or peaks exceeding 100 percent modulation of the RF signal. Its action shall eliminate noise spikes of one microsecond or less duration to prevent loss of intelligibility due to reduction in receiver gain under the influence of continuous RF noise such as are caused by commutators, ignition systems.

3.6.1.23.8 Cross modulation and desensitization. Cross modulation and desensitization effects of interfering signals (modulated 30 percent at 1,000 Hz) on channels more than one (1) MHz from and at a level 70 dB above the desired channel reference level (-103 dBm at 10 dB signal plus noise-to-noise) shall permit a signal plus noise-to-noise ratio of the desired signal of at least six dB for both main and guard receivers.

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3.6.1.23.9 Channel interference. For a -43 dBm unmodulated carrier signal on any channel with no signal on any other channel, the quieting measured on any other channel greater than 100 KHz away shall be not greater than the quieting produced on that channel for a signal of -110 dBm (squelch disabled) for both main and guard receivers.

3.6.1.23.10 RF intermodulation. For an RF input of two individual unmodulated carriers each of minus 43 dBm on any two channels, there shall be no quieting on any other channel greater than the quieting produced on that channel by a minus 110 dBm unmodulated carrier at that channel (squelch disabled) for both main and guard receivers.

3.6.1.23.11 Intermediate frequency rejection. The IF rejection of the main and guard receivers shall be as specified in Tables X and XI.

TABLE X. Main receiver IF rejection

Channel Frequency	1st IF Rejection (29 MHz)	2nd IF Rejection (455 KHz)
30-40 MHz	NLT 55 dB	NLT 90 dB
40-88 MHz	NLT 65 dB	NLT 90 dB
108-174 MHz	NLT 80 dB	NLT 90 dB
225-400 MHz	NLT 80 dB	NLT 90 dB

TABLE XI. Guard receiver IF rejection

Frequency	1st IF Rejection (28.045 MHz)	2nd IF Rejection (500 KHz)
40.5 MHz	NLT 80 dB	NLT 90 dB
121.5 MHz	NLT 80 dB	NLT 90 dB
156.8 MHz	NLT 80 dB	NLT 90 dB
243.0 MHz	NLT 80 dB	NLT 90 dB

3.6.1.23.12 Image and spurious rejection. Image rejection shall be not less than 80 dB relative to the response to the desired signal for the main and guard receivers except that the image frequency rejection of the 156.8 MHz guard receiver shall be not less than 60 dB. Spurious rejection of the main and guard receivers shall be not less than 70 dB relative to the response of the desired signal.

3.6.1.23.12.1 Internal spurious. Degradation of sensitivity of the main or guard receivers due to internal spurious shall be limited to no more than 100 channels out of the 11,960 channels available. In addition, no more than 50 of the 100 shall require a signal level of more than -97 dBm to meet a S + N/N ratio of 10 dB (see 3.6.1.23.1). The complete list of degraded channels shall be reported to the procuring activity and the frequencies, sensitivity level, and these channels shall be listed in handbooks and support documentation.

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3.6.1.23.13 AGC recovery and blocking. When signals up to +7 dBm are present, it shall be possible to obtain a 10 dB signal-plus-noise-to-noise ratio with -103 dBm input not later than 0.25 second after the one volt signal has been removed for both main and guard receivers. Blocking shall not occur on desired signals up to +7 dBm. Blocking is defined as a 3 dB or greater reduction of the 250 mW audio output level across 600 ohms or 150 ohms.

3.6.1.23.14 Input signal protection. The main and guard receivers shall not be damaged by inputs up to +27 dBm (on channel).

3.6.1.23.15 Regeneration. The interstage and overall regeneration of the main and guard receiver shall be minimized so that no evidence of instability occurs at maximum gain.

3.6.1.23.16 Receiver amplitude variation (FM). The limiting characteristics of the main and guard receivers shall be such that the audio output shall not vary more than 3 dB when the RF input signal deviated 2.4 KHz at 1 KHz FM is varied from the specified sensitivity level (see 3.6.1.23.1) up to +7 dBm.

3.6.1.23.17 Main receiver characteristics. The main receiver shall meet all the requirements of 3.6.1.23 through 3.6.1.23.16 plus the requirements of 3.6.1.23.17.1 through 3.6.1.23.17.8 when the radio is in the transmit or receive, transmit or receive plus guard or ADF modes.

3.6.1.23.17.1 Wideband audio outputs. The main receiver shall have one wideband audio output for use with secure voice systems and one for use with ADF systems. The outputs shall not pass through the squelch circuit and shall be operative at all times when in the receive mode. The ADF output shall be connected to the AM detector regardless of whether the receiver-transmitter is operating in AM or FM.

3.6.1.23.17.2 Wideband audio load impedance. The radio set shall provide specified output performance on the secure voice and ADF outputs when loaded with unbalanced load impedances between 2000 ohms and 20,000 ohms.

3.6.1.23.17.3 Wideband audio output levels. The TSEC/KY-28/58 system output level shall be not less than 0.250 volts rms to not greater than 8.3 volts rms across 20,000 ohms with inputs modulated 90 percent (AM) or deviated ± 8.0 KHz (FM) at 1,000 Hz. The ADF system output level shall be 4.0 volts peak to peak minimum across 20,000 ohms with an RF input of -53 dBm modulated 90 percent (AM) at 1,000 Hz.

3.6.1.23.17.4 Wideband audio output responses. The receiver wide band audio output for TSEC/KY-28/58 shall be within +2 to -4 dB of the 1,000 Hz reference level for modulating signals between 30 Hz and 25 KHz. The frequency response of the ADF audio shall be within ± 2dB of the 1,000 Hz reference level for modulating signals between 50 Hz and 10 KHz.

3.6.1.23.17.5 Wideband audio output distortion. Total harmonic distortion at the receiver wideband audio outputs shall be not greater than eight percent for 90 percent modulation (AM) or deviation of ± 8.0 KHz at 1,000 KHz FM with signal levels up to +7 dBm. Under service conditions, the distortion shall be not greater than 10 percent.

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3.6.1.23.17.6 Wideband audio output phase. The ADF audio output shall be not greater than 20 degrees out of phase with the negative portion of the RF signal modulation envelope when the RF signal is modulated 30 percent at 200 Hz. (Nominal phase delay = $180^{o} + 18 \times 10^{-3} F_{m}$ where F_{m} = ADF modulation frequency.)

3.6.1.23.17.7 Receiver FSK capability. In the wide band mode, the receiver shall be capable of demodulating FSK signals with data rates up to 10 KB/S with deviation up to ± 20 KHz. The following characteristics pertain to the wideband output.

a. Output level. 2VPP minimum into 600 ohms, unbalanced for ± 20 KHz deviation at bit rates from 50 B/S to 10 KB/S.

b. Phase. Positive going deviation from center frequency shall produce a positive going demodulated output.

c. Delay. 50 microseconds ± 15 microseconds at the 50 percent amplitude points.

d. Rise and fall times. No more than 20 microseconds between the 10 percent and 90 percent amplitude points.

3.6.1.23.18 Guard receiver characteristics. The guard receiver shall meet the applicable requirements of 3.6.1.23 through 3.6.1.23.16, 3.6.1.23.18.1, and 3.6.1.23.18.2 when in the transmit or receive plus guard, ADF, emergency, guard precedence or guard select modes. The requirements of 3.6.1.23.17 do not apply to the guard receiver. The guard receiver shall be capable of operation on fixed frequencies which are 40.5 MHz, 121.5 MHz, 156.8 MHz, and 243 MHz. The guard receiver shall be required to receive AM normal voice and pulsed tone at 121.5 MHz and 243.0 MHz. The guard receiver shall share the RF antenna input with the main receiver and FM normal voice and pulsed tone at 40.5 MHz and 156.8 MHz. Both the main and guard receiver shall be able to receive signals simultaneously when the receiver-transmitter is in the receive condition.

3.6.1.23.18.1 Low level guard audio output. A separate line shall be provided for a low level guard receiver output. The level shall be between 0.15 and 0.45 VRMS into a 600 ohm ± 10 percent resistive load at -53 dBm input level, 30 percent AM at 1 KHz or -53 dBm input level, ± 2.4 KHz FM at 1 KHz. This guard receiver output is for use with the TSEC/KY-28/58. With a reference input of -53 dBm, 30 percent modulated AM or ± 2.4 KHz FM at 1 KHz, the frequency response shall be within +1 to -3 dB from 500 to 3,500 Hz and distortion at 1,000 Hz shall be not greater than five percent. It shall be possible for this output level to be increased by 20 dB to a range of 1.5V to 4.5 VRMS by intermediate maintenance level personnel. Under service conditions, the distortion shall be not greater than 10 percent.

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3.6.1.23.18.2 Guard precedence operation. The radio set shall have provision for guard precedence frequency selection from an external switch. This mode which is intended for emergency use is activated by closing an external switch (not included in receiver-transmitter) to provide an electrical ground on an assigned pin. When this switch is closed the front panel controls are disabled, the radio is turned on; the main transmitter and receiver are tuned to the 243.000 MHz (AM) guard frequency. Activation of this mode shall take precedence over any other mode of operation selected.

3.6.1.24 Transmitter characteristics.

3.6.1.24.1 Unmodulated power output.

3.6.1.24.1.1 FM power. Under all service conditions except as specified below, the RF power output shall be not less than 15 watts over the frequency range of 30 to 87.975, 156 to 173.975 MHz, and 225 to 399.975 MHz. From 22 to 18 VDC input the RF power output may decrease from 15 watts to a minimum of 2 watts. From 55 degrees C to 71 degrees C ambient the RF power output may decrease 3 dB relative to power level at 55 degrees C. From 71 degrees C to 95 degrees C ambient the RF power output may decrease 6 dB relative to the power level at 55 degrees C.

3.6.1.24.1.2 AM power. Under all service conditions except as specified below, the RF power output shall be not less than 10 watts with an unmodulated carrier over the frequency range of 118.000 through 155.975 MHz and 225.000 through 399.975 MHz. From 22 to 18 VDC input the RF power output may decrease from a minimum of 10 watts to 2 watts minimum. From 55 degrees C to 71 degrees C ambient the RF power output may decrease 3 dB relative to the power level at 55 degrees C. From 71 degrees C to 95 degrees C ambient the RF power output may decrease 6 dB relative to the power level at 55 degrees C.

3.6.1.24.2 Transmitter RF termination. The transmitter shall meet the specific performance requirements with an RF termination of 50 ohms $\pm$ 3 ohms.

3.6.1.24.3 Transmitter output circuit. The transmitter shall operate as specified after being subjected to either an open or short circuit load impedance or any load impedance.

3.6.1.24.3.1 Mismatch load. The transmitter shall meet the requirements of 3.6.1.24.1.1 and 3.6.1.24.1.2 when terminated in the load of 3.6.1.24.2. At the worst case phase angle at a voltage standing wave ratio (VSWR) of 2.5:1, the power turndown shall be not greater than 2 dB. The transmitter shall turn down gradually for VSWR greater than 2.5:1 and the transmitter shall deliver not less than 2 watts forward power regardless of VSWR.

3.6.1.24.4 Tone transmissions.

3.6.1.24.4.1 Squelch tone. In the 30-88 MHz VHF (FM) band a squelch keying tone shall be transmitted along with the normal voice modulation. The tone shall be 150 Hz $\pm$ 2 percent and shall frequency modulate the transmitter output 3 KHz $\pm$ 500 Hz. The tone shall be inhibited during secure voice and AJ transmissions.

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3.6.1.24.4.2 Tone. The transmitter modulator circuit shall include a tone generator for modulating the transmitter with a 1.020 KHz ± 10 percent frequency on any channel above 118 MHz. The percentage of modulation shall be not less than 60 percent or FM deviation of ± 3.4 KHz. A separate connector pin shall be provided for tone activation. The tone shall be transmitted by grounding this pin.

3.6.1.24.5 Transmit precedence. Activating the transmit keyline shall cause the transmitter to transmit through the communications antenna port, J1, only.

3.6.1.24.6 Transmit key lines. Two pins shall be provided on the rear connector to control radio set transmissions. The transmit keyline (push-to-talk or PTT) shall cause transmitter output if the other keyline (transmit blanking) is open (not grounded). If the transmit blanking line is grounded, the transmitter shall be inhibited regardless of whether the PTT line is at ground or open.

3.6.1.24.6.1 RF carrier delay from PTT line. The RF carrier shall reach 90 percent of specified output level within 70 milliseconds after actuation of the push-to-talk (PTT) line.

3.6.1.24.6.2 RF carrier rise and fall time. With the PTT line grounded, the RF carrier shall rise to 90 percent of full power output within 200 microseconds after the transmit blanking line is ungrounded. The RF carrier shall fall to -80 dBc within 80 microseconds after the transmit blanking line is grounded.

3.6.1.24.7 Transmitter sidetone. In the AM mode (normal voice) a transmitter sidetone shall be developed by sampling a portion of the modulated transmitter carrier in the antenna circuit. The sidetone signal shall be introduced into the audio portion of the main receiver. In the FM (normal voice) mode the sidetone may be sampled from the modulation audio circuitry and gated with the forward power monitor.

3.6.1.24.7.1 Sidetone level. With a normal voice audio input voltage of 0.5 volt rms closed-circuit at 1,000 Hz, the sidetone audio output at the main receiver shall be not less than 250 milliwatts across 600 ohms when the volume control is in full clockwise position. An internal pot shall be provided to permit adjustment of the sidetone level to 3 dB below the standard level.

3.6.1.24.8 AM or FM inputs. Inputs shall be provided for normal voice, secure voice and retransmit. The normal voice input shall provide a screwdriver adjustment accessible from outside the radio set which permits adjustment of the normal voice input level such that a level between 0.25 volt rms and 1.5 volts rms (closed circuit) will produce 90 percent AM or ± 5.6 KHz FM deviation. The nominal factory setting shall be 0.5 volt rms.

3.6.1.24.8.1 Audio input impedances. The normal voice audio input impedance shall be 175 ohms ± 25 percent unbalanced. The secure voice audio input impedance shall be 2,000 ohms ± 25 percent unbalanced. The retransmit audio input impedance shall be not less than 1000 ohms ± 10 percent unbalanced.

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3.6.1.24.8.2 Amplitude modulation.

3.6.1.24.8.2.1 Percent modulation. Under standard conditions the percent modulation shall be linear with not less than 80 percent upward and between 90 and 100 percent downward for a 0.5 volt rms (closed circuit) 1000 Hz signal at the normal voice audio input, a 6.5 volt rms (closed circuit) 1000 Hz signal at the retransmit audio input, or a 12 volt peak-to-peak 16,000 bits per second closed circuit signal at the secure voice audio input. The average modulated power out shall be not less than 1.30 times the unmodulated power output. Under environmental conditions, the percent modulation shall not be less than 65 percent upward and between 80 and 100 percent downward. The average modulated power output shall be not less than 1.20 times the unmodulated power output for environmental conditions.

3.6.1.24.8.2.2 Carrier fidelity. With a constant normal voice audio input voltage of 0.5 volt rms (closed circuit) adjusted to any frequency between 500 and 3,500 Hz, the demodulated audio voltage shall be within +1 dB, -3 dB of the demodulated audio voltage resulting from a 0.5 volt (closed circuit) 1,000 Hz audio input. With a constant retransmit audio input voltage of 6.5 volts rms (closed circuit) adjusted to any frequency between 500 and 3,500 Hz, the demodulated audio voltage shall be within +1 dB, -3 dB of the demodulated audio voltage resulting from a 6.5 volt (closed circuit) 1,000 Hz audio input. With a constant secure voice audio input voltage of 12 $\pm$ 1 volts peak to peak (closed circuit) adjusted to any frequency between 16 and 10,000 Hz when the narrow band premodulation filter is used (see 3.6.1.24.8.4) and 300 and 20,000 Hz when the wideband premodulation filter is used, the demodulated audio voltage shall be within +1 dB, -3 dB of the demodulated output at 1,000 Hz modulation.

3.6.1.24.8.2.3 Distortion. With a normal voice audio input of 0.35 VRMS (closed circuit) or a retransmit audio input of 4.5 VRMS (closed circuit) adjusted to any frequency between 500 Hz and 3500 Hz, the demodulated audio harmonic distortion shall be not greater than five percent. Under service conditions the distortion shall be not greater than eight percent. From +71°C to +95° ambient temperature, the distortion shall be not greater than fourteen percent. With a secure voice audio input of 3.0 VRMS (closed circuit) adjusted to any frequency between 16 Hz and 10,000 Hz with the narrowband premodulation filter or between 300 Hz and 20,000 Hz with the wideband premodulation filter, the demodulated audio harmonic distortion shall be not greater than 10 percent, under standard or service conditions. From +71°C to +95°C ambient temperature, the demodulated audio harmonic distortion shall be not greater than sixteen percent.

3.6.1.24.8.2.4 Incidental frequency modulation. With the RF carrier amplitude modulated not less than 75 percent upward and 90 percent downward at 1,000 Hz, the peak frequency deviation arising from incidental frequency modulation (FM) of the transmitter shall be not greater than $\pm$ 2 KHz.

3.6.1.24.8.3 Frequency modulation characteristics. The frequency modulation of the RF output for the normal voice, secure voice and retransmit inputs, shall have the characteristics specified in 3.6.1.24.8.3.1 through 3.6.1.24.8.4.

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3.6.1.24.8.3.1 Deviation. An audio signal of 0.5 volts rms (closed circuit) applied to the normal voice audio input or an audio signal of 6.5 VRMS (closed circuit) applied to the retransmit audio input shall produce an FM modulated transmitter RF output signal with a deviation of ± 5.6 KHz ± 0.5 KHz. An audio signal of 12 V ± 1 V peak-to-peak (closed circuit) applied to the secure voice input shall produce an FM modulated transmitter output signal with a deviation of ± 5.6 KHz ± 0.5 KHz. Under service conditions, the deviation produced shall be not less than ± 4.5 KHz.

3.6.1.24.8.3.2 Audio frequency response. With a constant normal voice audio input voltage of 0.5 volt rms (closed circuit) adjusted to any frequency between 500 and 3500 Hz, the deviation shall be ± 5.6 KHz ± 1.2 KHz. With a constant retransmit audio input voltage of 6.5 volts rms (closed circuit) adjusted to any frequency between 500 and 3500 Hz, the deviation shall be ± 5.6 KHz ± 1.2 KHz. With a constant secure voice audio input voltage of 12 ± 1 volts peak-to-peak (closed circuit) adjusted to any frequency between 16 Hz and 10,000 Hz when the narrowband premodulation filter is used, and any frequency between 300 Hz and 20,000 Hz when the wideband premodulation filter is used, the deviation shall be ± 5.6 KHz ± 1.2 KHz.

3.6.1.24.8.3.3 Modulation distortion. With any audio input frequency specified in 3.6.1.23.8.3.2 for the normal voice, secure voice and retransmit inputs and with the audio input adjusted for ± 5.6 KHz deviation, the distortion shall be not greater than five percent. Under service conditions the distortion shall be not greater than eight percent.

3.6.1.24.8.3.4 Deviation capability. The radio set shall be capable of FM/FSK deviation up to ± 25 KHz for AJ applications.

3.6.1.24.8.3.5 Incidental AM. With maximum deviation (± 5.6 KHz) the amplitude modulation of the carrier shall be not greater than four percent.

3.6.1.24.8.4 Premodulation filters. Two filters shall be incorporated in the radio to shape the audio input prior to modulation and after the limiting bandwidths specified in Table XII.

TABLE XII. Filter bandwidth

Filter type	6dB	60dB
Narrowband Filter	18 + 0.5, -2.0 KHz	50 KHz max.
Wideband Filter	25 ± 2.0 KHz	100 KHz max.

The narrowband filter shall allow adjacent channel operation, with antenna isolation of 50 dB, between AN/ARC-182(V) radio sets in the normal and FM secure modes. The filters shall be selectable and shall operate in conjunction with the receiver selectivity filters such that when the receiver is in the narrowband mode the transmitter shall be in narrowband mode. When the receiver is in the wideband mode the transmitter shall also be in the wideband mode.

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3.6.1.24.10 Microphone bias voltage. The normal voice audio input shall provide bias voltage and current to operate an amplifier (AM-3597C/A type) for a dynamic microphone. The voltage supplied shall be 16 volts DC minimum with an external 1000 ohm resistive load. Under emergency power conditions, the voltage supplied shall be not less than 12 volts DC with an external 1000 ohm resistive load.

3.6.1.24.11 Transmit energy spectrum. The transmitter output shall meet the requirements of 3.6.24.11.1 through 3.6.1.24.11.4.

3.6.1.24.11.1 Transmit noise floor. The transmitter broadband noise level shall be no more than -120 dBm/Hz at any frequency 10 percent or more removed from the carrier. For carriers above 100 MHz, the 10 percent removal requirement is reduced to 10 MHz removal.

3.6.1.24.11.2 Spectral containment. Ninety-nine percent of normal and baseband secure voice transmitted energy shall be contained within the operating 25 KHz channel.

3.6.1.24.11.3 Transmitter spurious. Transmitter spurious outputs greater than 100 KHz from the carrier shall be not less than 66 dB below the carrier except for crossover spurious as specified in Table XIII.

3.6.1.24.11.4 Transmitter harmonics. The transmitter harmonics shall be not greater than the limits of Table XIV.

3.6.1.24.12 Noise modulation of transmitter.

a. AM. The noise on the unmodulated carrier shall be 40 dB below the detected audio voltage of a carrier modulation of 90 percent at 1 KHz, measured in a 3 KHz audio bandwidth.

b. FM. The noise of the unmodulated carrier shall be 30 dB below the detected audio voltage of a 1 KHz tone having a deviation of $\pm$ 5.6 KHz, measured in a 3 KHz audio bandwidth.

3.6.1.24.13 Transmitter keying. When operating in normal or secure voice modes the transmit key line shall key the transmitter when voltages from ground to $\pm$ 5V are applied. An open circuit or above 12 VDC on the key line shall cause the radio to operate in receive mode.

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TABLE XIII. Transmitter spurious limits

Crossover Center Frequency (MHZ)	Frequency Range Of Crossover (MHz)	DF Range OF Crossover (MHz)	Spurious Level dB Below Carrier (DBC)
34.600	34.450 to 34.750	± 0.3	-50
40.800	40.650 to 40.950	± 0.5	-50
57.450	57.350 to 57.550	± 0.6	-50
66.075	65.550 to 65.875	± 1.6	-50
	65.900 to 66.200	± 0.5	-40
	66.225 to 66.450	± 1.6	-50
68.000	67.550 to 67.825	± 1.4	-50
	67.850 to 68.150	± 0.25	-40
	68.175 to 68.450	± 1.4	-50
81.800	81.600 to 82.000	± 0.5	-50
134.075	133.600 to 133.875	± 1.4	-50
	133.900 to 134.200	± 0.4	-40
	134.225 to 134.450	± 1.4	-50
136.000	135.500 to 135.825	± 1.6	-50
	135.850 to 136.150	± 0.25	-40
	136.175 to 136.500	± 1.6	-50

Notes:

1. Crossover center frequency is where the crossover spurious lie exactly on or very close to the carrier frequency.

2. Frequency range of crossover refers to the channel frequencies where the crossover pair of spurious exceed -66 DBC.

3. DF range of crossover specifies the maximum spacing from carrier where the pair exceed -66 DBC.

4. The spurious level specifies the maximum level of the crossover pair in the frequency range.

TABLE XIV. Transmitter harmonic limits

Frequency Band	Harmonic Level (DBC)					
	2ND	3RD	4TH	5TH	6TH	7TH
30-88 MHz	-55	-55	-66	-55	-66	-66
118-174 MHz	-55	-55	-62	-55	-66	-66
225-400 MHz	-55	-55	-55	-66	-66	-66

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3.6.1.25 Automatic relaying operation. In the T/R or T/R PLUS guard mode the main receiver and transmitter of the radio set shall operate in an automatic relaying mode (both normal and secure voice) when used in conjunction with another AN/ARC-182(V) or AN/ARC-159(V) radio set plus C-11628/ARC control, radio set and SA-2498/ARC switching unit or other appropriate external interface switching circuitry. The set shall provide on pins of the main power connector, the following inputs, outputs, and controls that allow the radio set to be used for automatic relaying in the transmit/receive or transmit/receive plus guard mode:

a. Main receiver audio outputs

b. Retransmit audio inputs

c. Relay trigger voltage (squelch indication)

d. Transmit control (T/R control line)

The application to the antenna circuit of either radio set, of a signal on the receiver selected frequency, and of a level sufficient to open the receiver squelch circuit shall cause the other radio set to automatically generate and emit a radio frequency carrier wave of the units rated power on its selected frequency. Removal of the actuating receiver signal shall cause the cessation of the generated carrier. When the actuated signal is 3 dB or more above the level required for relaying as described above, the generated carrier shall reach 90 percent of rated power level within the receiver and transmitter operate times. The system shall return to standby conditions after removal of the actuating signal. The receiver-transmitter shall perform relay function on any two RF channels in any of the frequency bands (see 3.6.1.1) provided a frequency separation of at least 10 MHz exists between the two channels with an isolation of 45 dB between the two antennas, and the spurious frequencies are not selected as relay channels.

3.6.1.26 TSEC/KY-28/58 operation. In the T/R or T/R+G mode the radio set shall operate in conjunction with the TSEC/KY-28 or TSEC/KY-58 speech security equipment described in publication CSEEB-13, CSESD-14, and CSEEB-32 respectively. Specific radio circuits required for secure voice operation in the AM or FM mode shall be implemented by actuating two designated pins as required on the radio connector. When required (for diphase secure operation) actuating one pin will switch the transmitter premodulation filter and the receiver IF to the wide bandpass. In AM mode the receiver-transmitter shall be capable of baseband and diphase operation; in FM mode-baseband only. In the FM mode the baseband secure voice signals shall be contained in the 25 KHz channel.

3.6.1.27 Maritime offset. The receiver-transmitter shall provide half-duplex operation in selected channels in the maritime band when a pin on the rear connector (J2) is grounded. The frequency which is displayed in any form in the system shall be the frequency of the transmitter. The receiver frequency shall be $\pm$ 4.6 MHz offset from the transmit frequency dependent on the transmit frequency chosen. Table XV lists the maritime offset channels of the receiver-transmitter available when the pin is grounded. If the pin is not grounded, the transmitter and receiver are on the same frequency.

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TABLE XV. Maritime offset frequencies

Transmit Frequency 1/ (MHz)	Receiver Frequency Offset (MHz)
156.000	0
156.025 through 156.275	+4.6
156.300	0
156.325	+4.6
156.350	+4.6
156.375 through 156.875	0
156.900 through 157.425	+4.6
157.450 through 160.600	0
160.625 through 160.875	-4.6
160.900	0
160.925	-4.6
160.950	-4.6
160.975 through 161.475	0
161.500 through 162.025	-4.6
162.050 through 173.975	0

1/ The transmit frequency is displayed during half-duplex operation.

3.6.1.28 Mode indications. All circuits connected to pins for indication or external control purposes shall be capable of pulling in a 100 milliampere external relay when a ground indication is provided. The indicator circuit shall withstand open circuit voltages up to 30 VDC and have leakage current of no more than 150 microamperes.

a. AM/FM mode indicator. A ground (≤1 VDC) shall be provided when the receiver-transmitter is operating in AM. An open circuit shall be provided for FM operation.

b. ADF mode indicator. A ground (≤1 VDC) shall be provided when the receiver-transmitter is operating in the DF mode at frequencies in the range of 108-173.975 MHz and 225-399.975 MHz. An open circuit shall be provided for all other operating modes.

c. UHF band indicator. A ground (≤1 VDC) shall be provided when the receiver-transmitter is operating in the frequency range of 225-399.975 MHz. An open circuit shall be provided for all other modes.

d. Low VHF-FM band indicator. A ground (≤1 VDC) shall be provided when the receiver-transmitter is operating in the frequency range of 30-87.975 MHz. An open circuit shall be provided for all other bands.

e. Retransmit key. A ground (≤1 VDC) shall be provided when the S+N/N ratio of the receive signal exceeds the squelch break threshold. An open circuit shall be provided when the receiver resquelches.

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3.7 Safety. The receiver-transmitter safety requirements shall be in accordance with MIL-E-5400, except resistance from ground pin to chassis shall be not greater than 20 milliohms.

3.7.1 Human performance/human engineering. MIL-STD-1472 and MIL-H-46855 shall be used as a design guide for the receiver-transmitter and all component parts therein.

3.8 Bonding. Bonding shall be in accordance with MIL-B-5087.

3.9 Workmanship. Workmanship for the receiver-transmitter shall conform with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Standard conditions. Unless otherwise specified herein, all inspections shall be performed with conditions specified as follows:

a.	Temperature	Room ambient (25 $\pm$ 10°C)
b.	Altitude	Normal ground

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c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power	28 ± 0.5 volt dc

4.4 First article inspection. First article inspection shall be conducted by the contractor on receiver-transmitters representative of the production receiver-transmitters to be supplied under the contract. First article tests shall be accomplished under the authorized test procedures (see 4.6). No first article inspections shall be conducted prior to acceptance of the first article test procedure by the procuring activity. First article inspection shall consist of all tests described in 4.4.1.

4.4.1 Scope of first article tests. First article tests shall include all tests in the approved test procedures to determine that the receiver-transmitter meets all the requirements of this specification and the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-T-5422 including temperature shock, a random vibration test as specified in 4.4.4, and an electromagnetic compatibility test in accordance with MIL-STD-461 and MIL-STD-462, as modified in 4.4.3. The first article test may also include a reliability qualification test.

4.4.2 First article approval. Approval of the first article sample will be by the procuring activity upon completion of all tests.

4.4.3 Electromagnetic interference tests. Compliance with the requirements of 3.4.4.3 shall be demonstrated by tests. These tests shall be performed in accordance with MIL-STD-462.

4.4.3.1 EMI test plan. The contractor shall perform EMI tests in accordance with an EMI Test Plan. The Test Plan shall describe all EMI tests required to demonstrate compliance with the requirements of 3.4.4.3, and shall address all anomalies, inconsistencies, and omissions (if any). The Test Plan shall detail a combination of operating modes and frequencies to be tested. The combination must represent all operating modes of the receiver-transmitter and at least four frequencies. The Test Plan shall be acceptable to the procuring activity prior to the commencement of EMI testing.

4.4.3.2 Ground plane interference. The contractor shall devise a procedure to demonstrate that the receiver-transmitter complies with the requirements of 3.4.4.3.4. The procedure shall be included in the EMI Test Plan.

4.4.3.3 Conducted susceptibility test, interconnecting and signal leads. In addition to the requirements of MIL-STD-461 the following RF susceptibility test shall be conducted. No malfunction, undesirable response, or change in indication beyond the tolerance given in the equipment specification or approval test plan shall be produced in any equipment when the RF signal levels listed in Table XVI are coupled into the interconnecting wiring.

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TABLE XVI. Susceptibility test signals

Frequency	Required Signal Levels In One Turn Loop
2 to 30 MHz	100 MV RMS

The RF signal shall be modulated 80 percent with 400 Hz and with that frequency and type of modulation to which the test sample is most susceptible. This requirement applies to all interconnecting wires or cable between component parts of subsystems, and to wires and cables connected to other equipment. The RF signal shall be measured in the one-turn loop with an RF voltmeter as shown on Figure 7. The signal generator shall have an output impedance of 50 ohms or less and a maximum open-circuit voltage of not less than 3 volts rms. The test procedure shall be as follows:

a. With the current probe clamped around the wire bundles under test, set the signal generator for maximum output and slowly scan from 2 MHz to 30 MHz.

b. At those frequencies where a response is noted, move the current probe along the wire for maximum indication. Reduce the signal generator output and determine the threshold of susceptibility of the wire bundle. When a response is found on an interconnecting wire bundle, probe each wire in the bundle separately to determine the susceptible wire(s).

c. Record and include in the test report the threshold of susceptibility data (frequency, amplitude, and description of responses) obtained in (b) above (see 6.2.2). Also include in the test report the corrective action required for each wire that does not meet the required level.

4.4.3.3.1 Wire bundles. Wire bundles may be tested in sections if the current probe will not fit around the complete wire bundles. Chassis or circuit grounds not normally routed with the interconnecting wiring shall be measured individually and shall not be included in the wire bundle under test. Power ground return leads shall not be placed in the probe.

4.4.4 Random vibration test. A random vibration test shall be conducted as specified in 4.4.4.1 through 4.4.4.4.

4.4.4.1 Test item operation. The test item shall be operated during application of random vibration so that the functional effects caused by these tests may be evaluated. The test item shall meet performance requirements, as specified, while the functional vibration levels are being applied and immediately preceding and following the application of the endurance levels.

4.4.4.2 Mounting techniques. The test item shall be hard mounted either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize effects of unbalanced loads. The input control

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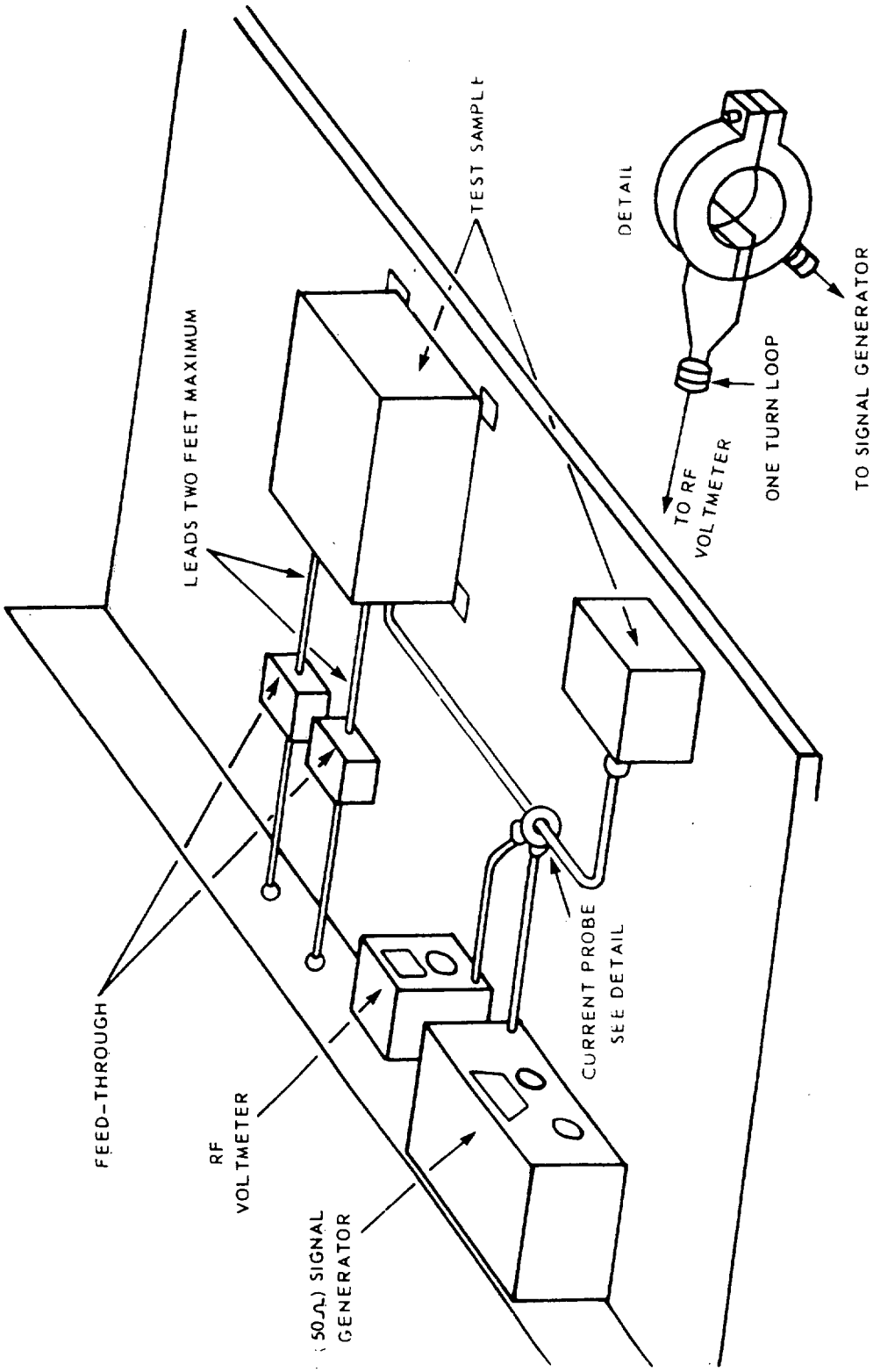


FIGURE 7. Conducted susceptibility test set up

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sensing device(s) shall be rigidly attached to the vibration table, or fixture if used, as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located in or on the test item to determine resonant frequencies and amplification factors. Locations to be selected should include main structure, printed circuit boards, large components, and modules, where practicable. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal.

4.4.4.3 Performance of test. The individual equipment test item shall be subjected to broadband random vibration excitation with the power spectral density envelope shown on Figure 8 for the endurance level test. The functional test envelopes for the vertical, longitudinal, and lateral direction are shown on Figures 9 and 10 respectively. The test item shall be attached to the vibration exciter according to 4.4.4.2. Vibration shall be applied sequentially along each of three orthogonal axis of the test item. For each axis of vibration, the appropriate functional vibration envelope for that axis shall be applied for 1/2 hour, followed by one hour of the endurance level vibration as shown on Figure 8, and finally another 1/2 hour of the previous functional vibration level shall be applied.

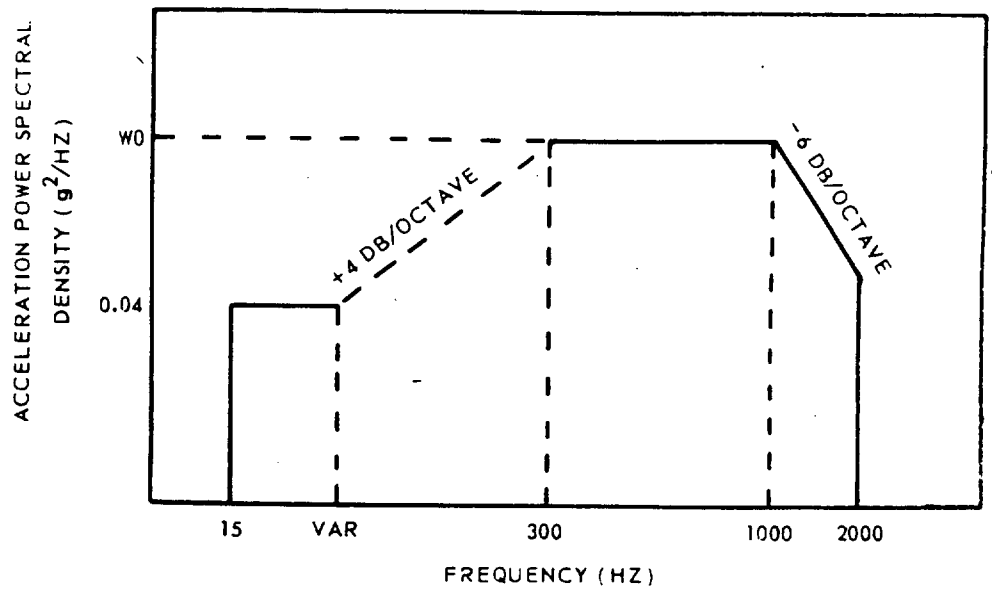


FIGURE 8. Random vibration envelope

The acceleration power spectral density (g^2/Hz) of applied vibration, as measured on the test fixture at mountingpoints of the test item, shall be as specified in 4.4.5.4. Test times shall, for each axis, be one hour each for functinal and endurance levels. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100, -30 percent (+3, -1.5 dB) below 500 Hz and +100, -50 percent (+3 dB) between 500 Hz and 2,000 Hz except that deviations as large as +300, -75 percent (±6 dB) shall be allowed over a

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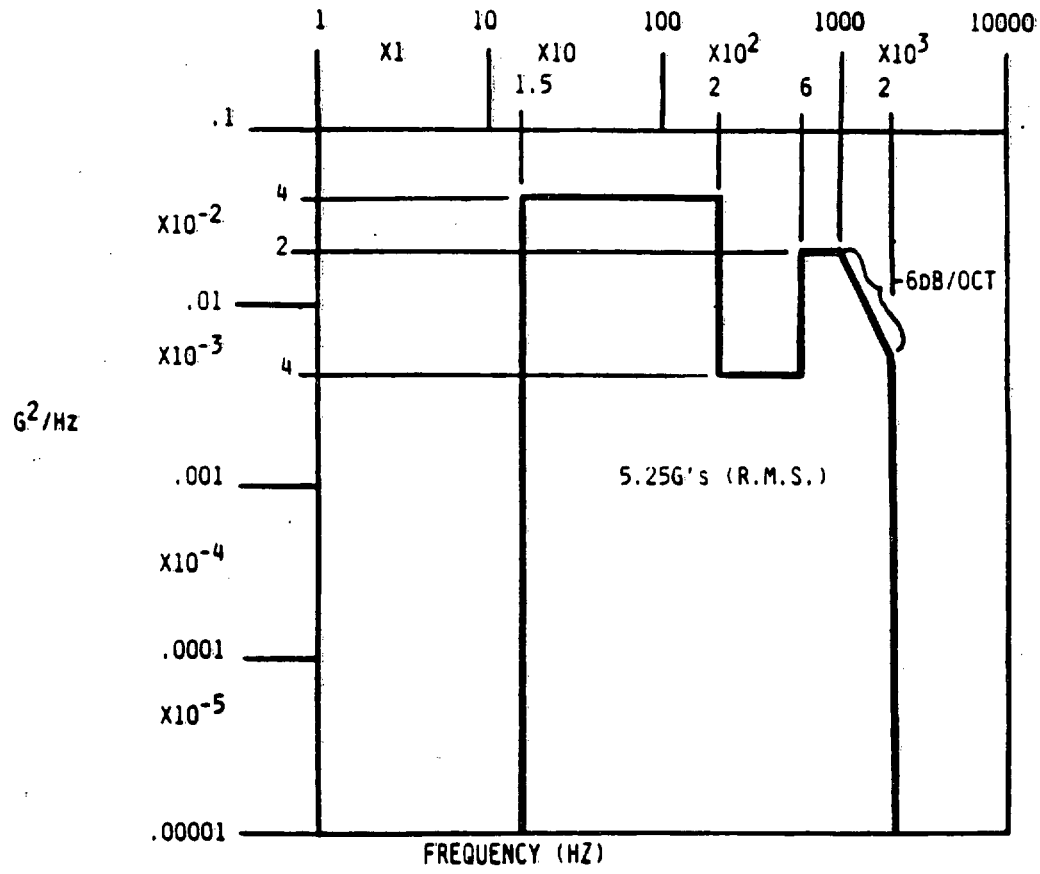


FIGURE 9. RT-1324A Random Vibration - Longitudinal & Lateral Planes (Functional)

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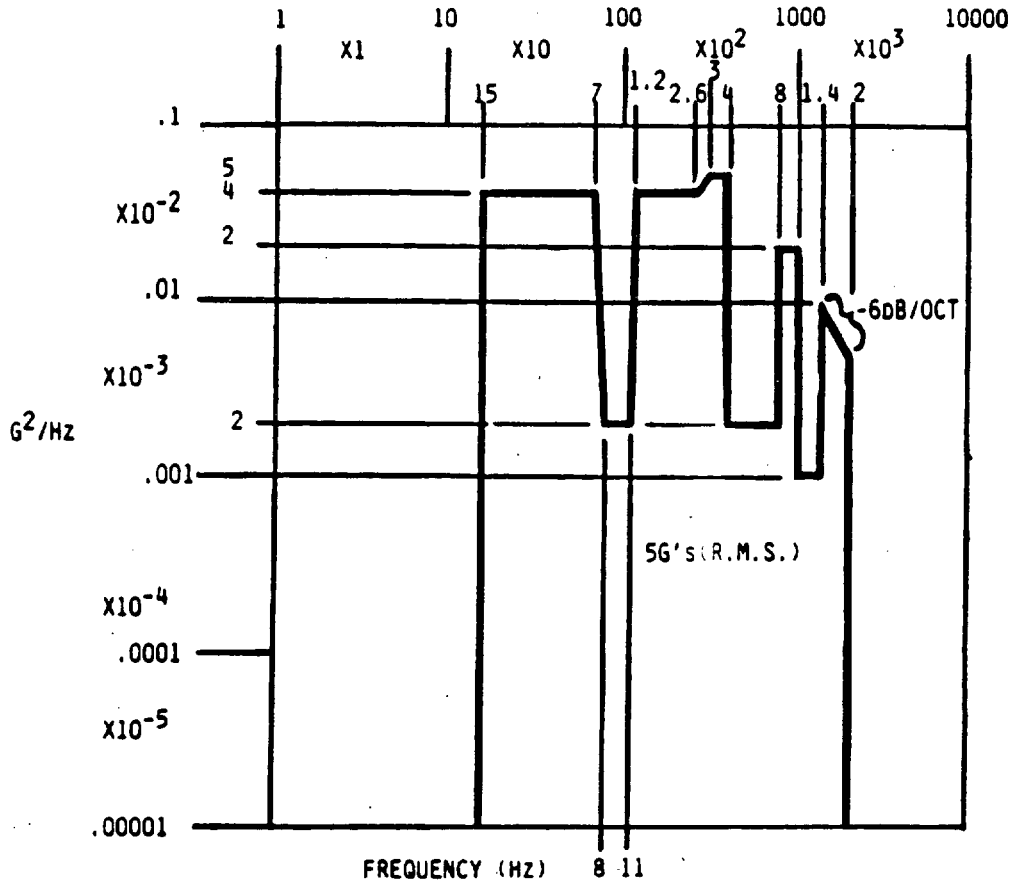


FIGURE 10. RT-1324A Random Vibration - Vertical Plane (Functional)

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cumulative bandwidth of 100 Hz maximum, between 500 and 2,000 Hz. Tolerance levels in terms of dB are specified as:

$$dB = 10 \text{ Log}_{10} \frac{W1}{WO}$$

Where W1 = measured acceleration power spectral density in g^2/Hz units, the term WO defines the specified level in g^2/Hz units. Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product, BT=50 minimum. Specific analyzer characteristics shall be as specified below or equivalent, subject to the BT=50, minimum limitation.

a. On-line, contiguous filter, equalization analysis system having a bandwidth as follows:

B=25 Hz, maximum between 20 and 200 Hz
B=50 Hz, maximum between 200 and 1,000 Hz
B-100Hz, maximum between 1,000 and 2,000 Hz

b. Swept frequency analysis systems characterized as follows:

(1) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B=25 Hz, maximum between 20 to 200 Hz
B=50 Hz, maximum between 200 to 1,000 Hz
B=100 Hz, maximum between 1,000 to 2,000 Hz

(b) Analyzer averaging time =T= 2 RC = 1 second, where T = true averaging time constant

(c) Analysis sweep rate (linear) =

$$R = \frac{B}{4RC} \text{ or } \frac{B^2}{8} \quad \text{(Hz/second)}$$

(2) Constant percentage bandwidth analyzer

(a) Filter bandwidth - pfc=one-tenth of center frequency maximum (0.1 fc) where p=percentage and fc=analyzer center frequency

(b) Analyzer average time = $T = \frac{50}{pfc}$ minimum.

(c) Analysis sweep rate (logarithmic) =

$$T = \frac{pfc}{4RC} \text{ or } \frac{(pfc)^2}{8}$$

(Hz/second), maximum, whichever is smaller

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Digital power spectral density analysis employing quantization techniques providing accuracies corresponding to the above approach.

4.4.4.3.1 Worst case axis. During the vibration testing, data shall be obtained to identify the worst case axis for purpose of vibration screening tests on production equipments.

4.4.4.4 Test levels. For the purpose of the random vibration test, the following test levels shall be used:

a. Functional Level-W_o = 0.05 g^2/Hz (8.5g rms minimum)

b. Endurance Level-W_o = 0.12 g^2/Hz (12.7g rms minimum)

4.4.5 Reliability qualification test. A reliability qualification test may be included in the first article tests. A minimum of six receiver-transmitters shall be tested in accordance with MIL-STD-781 using the reliability qualification test - mission profile in this specification. (See Figures 10 and 11.) Each receiver-transmitter shall have successfully completed the normal burn-in and all equipment test (AET) prior to the start of RQT. The mission profile test cycle should begin at the "12 hour" point which is a cold day ground stabilization (-54°C) prior to ground operation and flight in a cold day. Random vibration should be applied with the units hard-mounted (without vibration isolators). The accept-reject criteria imposed shall be those of MIL-STD-781, Test Plan IIIC, with the upper test MTBF specified in 3.3.1.3.

4.4.6 Maintainability demonstration. A maintainability demonstration for intermediate level requirements shall be conducted in accordance with MIL-STD-2084 and MIL-STD-471. Test Method 1, Test Plan B of MIL-STD-471, shall be used.

4.4.7 Production receiver-transmitters. Receiver-transmitters supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be capable of successfully passing the same tests as specified herein.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing may be under the supervision of the customer representative. The contractor shall retain test data showing quantitative results for all quality conformance tests. Such tests shall be signed and stamped by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

a. Individual tests (see 4.5.1)

b. Manufacturing screening (see 4.5.2)

c. Sampling tests (see 4.5.3)

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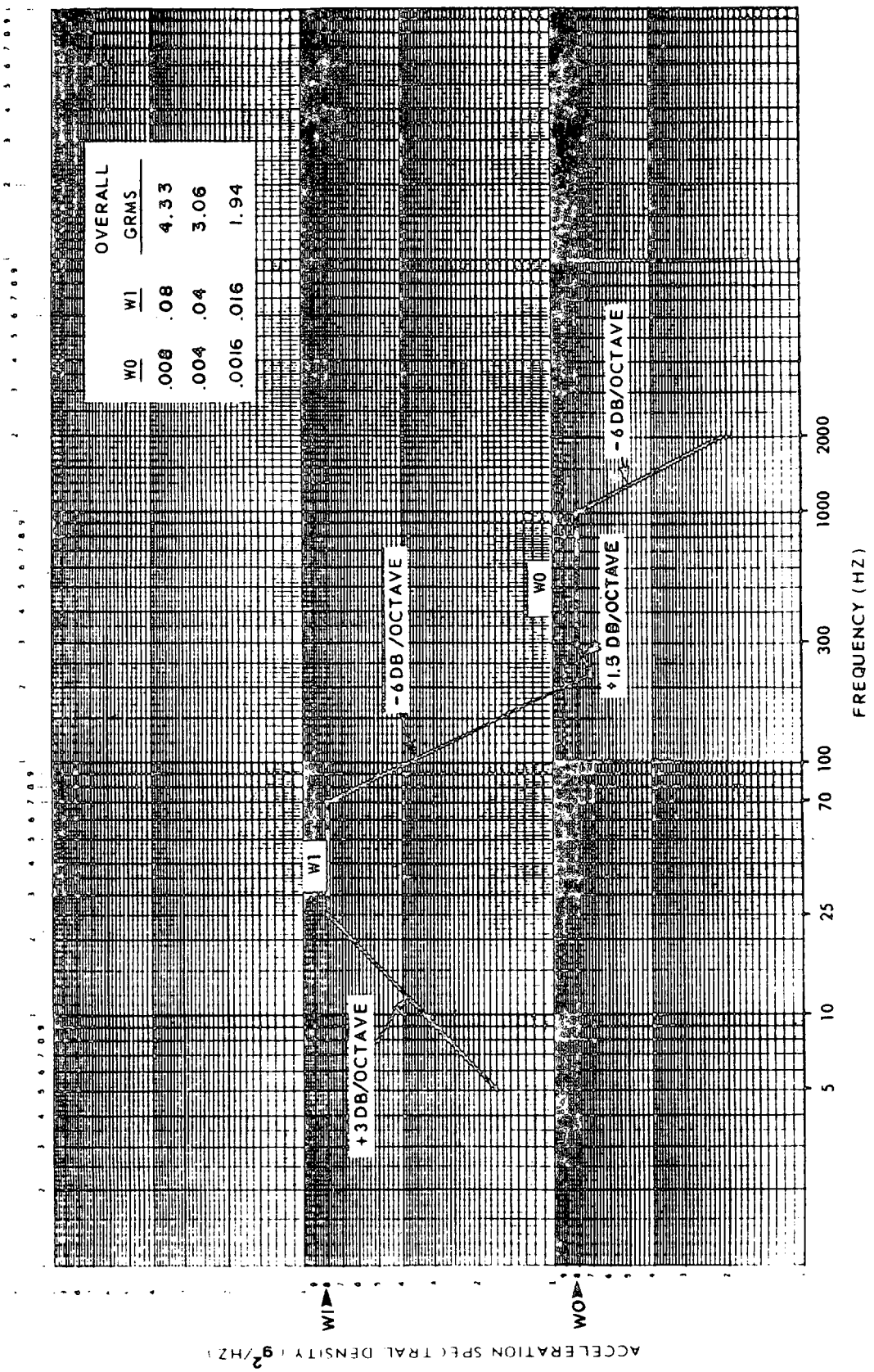


FIGURE 11. RQT vibration envelope (see 4.4.5)

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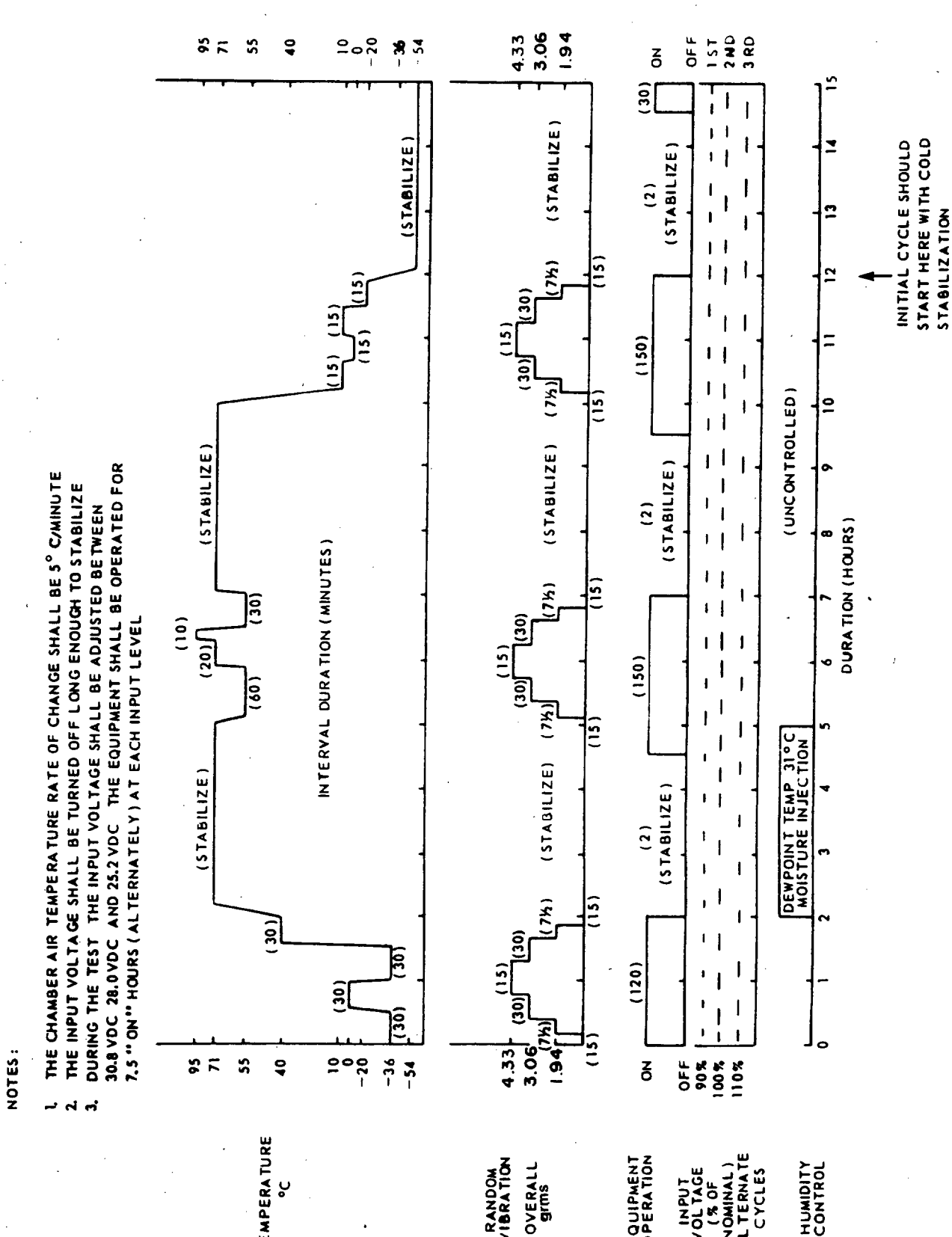


FIGURE 12. RQT mission profile test cycle (see 4.4.5 and 4.5.4.1)

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d. Reliability assurance tests (see 4.5.4)

e. Special tests (see 4.5.5)

4.5.1 Individual tests. Each receiver-transmitter submitted for acceptance shall be subjected to the individual tests. The tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each receiver-transmitter accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

4.5.1.1 Examination of product. Each receiver-transmitter shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each receiver-transmitter shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure specified equipment operation.

4.5.2 Manufacturing screening. The contractor shall subject each WRA to manufacturing screening prior to undergoing sample tests or reliability assurance tests. Manufacturing screening shall consist of WRA random vibration and temperature cycling. An additional screening period may be used at the option of the contractor. If an additional screening period is to be used, the details thereof must be included in the approved test procedures and must be the same for all equipment, including spare units.

4.5.2.1 WRA random vibration. Prior to conducting WRA temperature cycling, each WRA shall have completed a 10 minute failure-free vibration screen at 6.1g rms . Power spectral density shall cover 20-2000 Hertz. Pseudo-random vibration methods are acceptable. Vibration envelope is shown on Figure 12. The test item shall be hard mounted without vibration isolators and shall be operating throughout the test.

4.5.2.2 WRA temperature cycling. Each WRA shall undergo 25 temperature cycle between -54°C and 71°C. The rate of temperature change between extremes shall be at least 5°C per minute. When the WRA temperature has stabilized within 5°C or either temperature extreme, the WRA shall be soaked. Hot soak at +71°C shall be two hours; cold soak at -54°C duration shall be one hour. At the conclusion of each soak, the next temperature ramp shall begin. The WRA shall be energized throughout temperature cycling except during the cooldown period from +71°C to -54°C and during the 60 minute soak at -54°C. In addition, the WRA shall be turned off for one minute at points 30, 60, and 90 minutes into each hot soak period. Receiver-transmitter performance shall be periodically checked during each cycle and repairs shall be made to correct failures. The last five consecutive temperature cycles shall be failure free. One temperature cycle is defined as: operation at +71°C for the 120 minute hot soak (including three minutes "off"), non-operational cooldown to -54°C, 60 minute non-operational cold soak at -54°C, operational temperature increase to +71°C.

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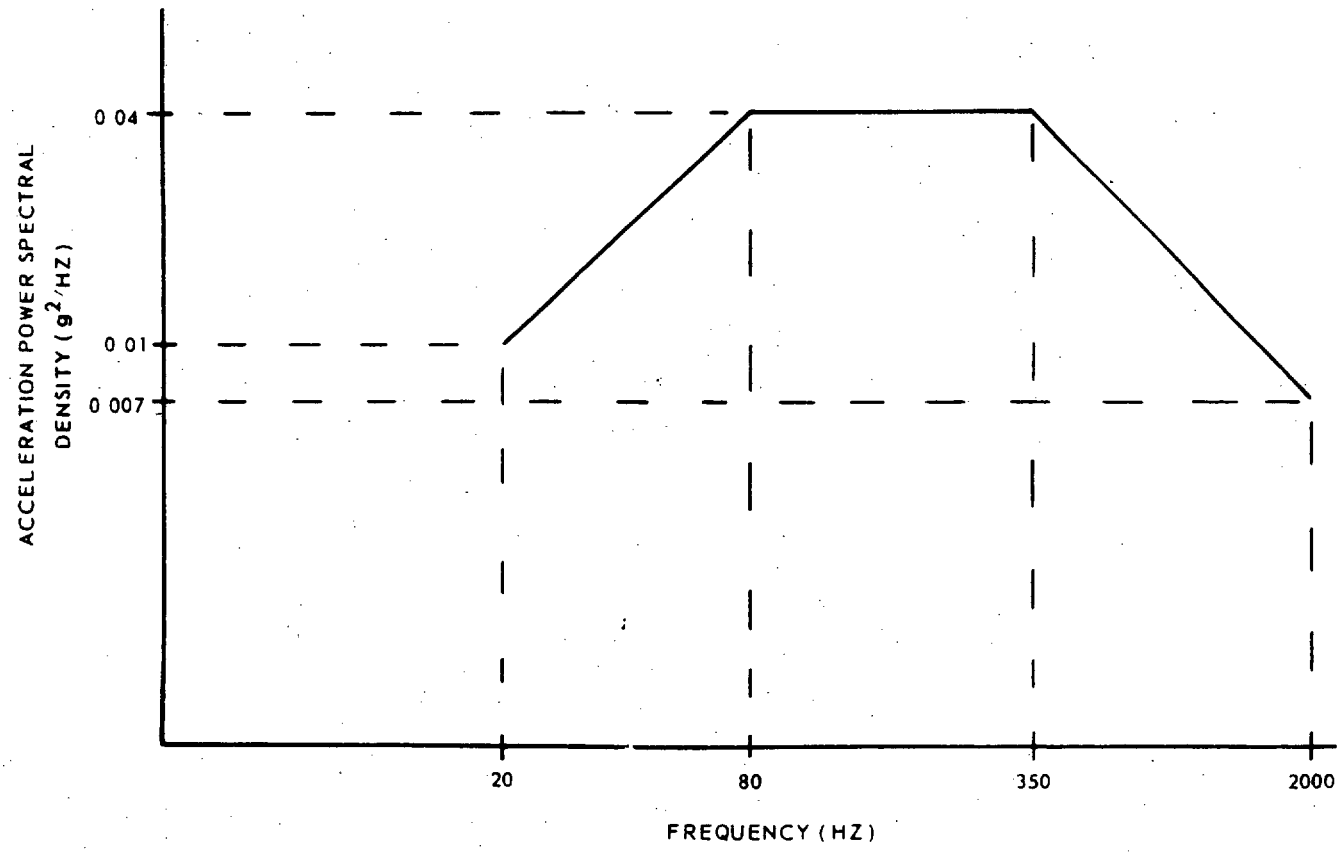


FIGURE 13. Manufacturing screening random vibration envelope (see 4.5.2.1)

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4.5.3 Sampling tests. Receiver-transmitters shall be subjected to sampling tests. Receiver-transmitters selected for sampling tests shall first have passed the individual tests. Receiver-transmitters will be selected for sampling tests by the government inspector in accordance with Table XVII.

TABLE XVII. Sampling test sample selection

Quantity of equipments offered for acceptance	Quantity to be selected for sampling test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling tests are not required when reliability assurance tests are conducted.

4.5.3.1 Scope of sampling tests. As a minimum, each receiver-transmitter selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each receiver-transmitter tested and should be based on results of the first article, individual and special tests.

c. Reliability sample test specified in 4.5.4.2. Each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurring in future equipment taken.

4.5.4 Reliability assurance tests. A reliability assurance test shall be conducted using MIL-STD-781. Classification of failure shall be in accordance with 4.5.7.

4.5.4.1 All equipment test. Each receiver-transmitter shall be tested as specified in MIL-STD-781 and modified herein. Temperature cycling and power on/off cycling shall be as shown on Figure 11 and vibration as shown on Figure 10 are required. The equipment shall be subjected to 4.5 hours of random vibration with equipment operating prior to the start of the temperature cycling and 4.5 hours of random vibration with equipment operating between temperature cycles 6 and 13. Each 4.5 hour random vibration period shall consist of three consecutive 90 minute cycles with three stepped levels as shown on Figure 10 and durations as shown on Figure 11. Random vibration may or may not be performed simultaneously with temperature cycling. Each receiver-transmitter shall have successfully completed manufacturing screening as specified in 4.5.2 prior to the start of the test. Each equipment (WRA) shall be tested for 125 equipment operating hours with the last two cycles

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failure free and in addition, the last 4.5 hours of random vibration shall be failure free. For determining MTBF compliance each equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The test MTBF shall be as listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract, the operating time and the failures therein (not counting any manufacturing screening failures or operating time) shall be totaled and the results compared with the reject line of Test Plan XVIIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any time the experience from the beginning of the contract is included. At any time that the current totals of test hours and test failures plotted on Test Plan XVIIIC curves show a right situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of receiver-transmitter at any time that a reject situation exists pending a review of the contractors efforts to improve the equipment, the equipment parts, the equipment workmanship, so that the entire compilation will show other than a reject decision.

4.5.4.2 Reliability sample test. When specified in the contract, (see 6.2.1) a monthly sample test of equipments shall replace the reliability test of 4.5.4.1. Units shall be tested as specified in MIL-STD-781 and modified herein. Temperature cycling and power on/off cycling are required and shall be as shown on Figure 11. Each sample equipment (WRA) shall be tested for 256 hours with the last two cycles failure free. For determining MTBF compliance, each designated equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The MTBF for each designated equipment shall be those listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures therein (not counting any burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan IIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. The sample size shall be based on the production quantity of that equipment for the month as specified in Table XVIII.

TABLE XVIII. Reliability test sample size

Production Quantity	Sample Size
2 - 5	1
6 - 25	2
26 - 50	3
51 - 100	4

4.5.4.3 Test details. Test details such as the performance criteria to be measured, special failure criteria, and preventive maintenance to be allowed during the test shall be part of the test procedures to be prepared prior to the beginning of the reliability assurance tests.

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4.5.5 Special tests. When specified in the contract, (see 6.2.1) special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the receiver-transmitter and to assure quality control. The receiver-transmitter selected for special tests may be selected from receiver-transmitter previously subjected to the sampling or reliability tests.

4.5.5.1 Special test schedule. Selection of receiver-transmitters for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.5.2 Scope of special tests. Special tests shall consist of such tests as authorized by the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and make it available to the procuring activity for acceptance prior to conducting the tests.

4.5.6 Equipment failure. Should a failure occur during either the sampling, reliability qualification, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure and the equipment BIT performance in detecting and isolating the failure.

b. Determine if the failure is an isolated case or design defect or pattern failure.

c. Prepare proposed corrective action intended to reduce the possibility of the same component or BIT failure(s) occurring in future tests or documented justification where no corrective action is proposed.

d. Where practical, include a test in the individual test to check all receiver-transmitters for this requirement until assurance is obtained that the defect has been corrected.

4.5.7 Failure classification. All incidents occurring during reliability assurance tests shall be classified and reported as either relevant or non-relevant in accordance with MIL-STD-2074 and MIL-STD-781. Only those incidents classified as relevant failures shall be accountable for making an accept or reject decision.

4.5.7.1 Relevant failures. All test incidents shall be considered as relevant failures unless determined to be non-relevant in accordance with the criteria of MIL-STD-2074.

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4.5.7.2 Non-relevant failures. Although non-relevant failures are not used for reliability assurance test accept or reject decisions, all test article incidents and equipment failures shall be recorded and reported.

4.6 Test procedures. The procedures used for conducting all tests required by Section 4 shall be prepared by the contractor (see 6.2.2). The right is reserved by the procuring activity or the government representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of the specification or the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures.

4.7 Testing of spares and repair parts. Equipment delivered as spare or repair parts shall undergo acceptance tests to ensure that the items meet performance and reliability requirements not less than those imposed in production.

4.7.1 Weapons replaceable assemblies. All items to be delivered as spares at the weapons replaceable assembly (WRA) or higher level shall receive performance, manufacturing screening and reliability assurance tests commensurate with that afforded the original production equipment.

4.7.2 Shop replaceable assemblies. All items to be delivered as spares at the shop replaceable assembly (SRA) or sub-SRA level shall receive functional checks and a manufacturing screening under conditions and for a duration commensurate with that imposed on that assembly in the original production.

4.7.3 Repair parts. All electronic component parts to be delivered as repair parts shall receive functional and screening tests equal to those imposed for the nearest military standard part.

4.7.4 Test details. The details of acceptance tests to be performed on SRA's and WRA's supplied as spare or repair parts shall be incorporated into item specifications, drawings and test procedures.

4.8 Presubmission testing. No item, part, or complete equipment shall be submitted until it has been previously tested by the manufacturer and found to comply, with all applicable requirements of this specification and the contract.

5. PACKAGING

5.1 General. All major units and parts of the receiver-transmitter shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555 for the level of shipment specified in the contract or purchase order (see 6.2.1).

5.2 Marking. Shipments shall be marked in accordance with requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. The receiver-transmitters specified herein are one portion of the AN/ARC-182(V) radio system. The AN/ARC-182(V) is a variable configuration of units using common modules that can be used to meet the prime and back-up UHF voice plus prime UHF/VHF/AM/FM voice and data communication requirements in any tactical aircraft. The variable configurations are intended to provide direct physical replacement of AN/ARC-159(V) radios. They are intended for production incorporation in new aircraft plus retrofit in selected aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. If a first article inspection is required (see 3.2, 6.3)

c. If non-standard parts or materials are required (see 3.3.5)

d. If reliability program is not to be established by contractor (see 3.4.1)

e. If receiver characteristics for main and guard receivers are not to be the same (see 3.6.1.3.c)

f. Whether All Equipment Test or Reliability Sample Test is required (see 4.5.4.1, 4.5.4.2)

g. If Special Tests are required (see 4.5.5)

h. Test procedure modifications or additional tests, if required (see 4.6)

i. Selection of applicable levels of packaging and packing (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Subpart 27.410-6 (DD Form 1423) are invoked, and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

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Paragraph	Data requirement title	Applicable DID no.
3.3.2	Engineering Change Proposals (ECP's) and Requests for Deviations and Waivers (Long Form)	DI-E-2037
3.4.1, 4.5.6.c	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719
3.3.3.	Program Parts Selection List (PPSL)	DI-MISC-80072
3.4.1.2	Part Approval Request	DI-MISC-80071
3.6.1.15, 4.5.5.2, 4.6	Production/Acceptance Inspection Procedures	DI-T-4903
3.6.1.23.12.1, 4.4.1, 4.4.3	First Article Inspection Report	DI-T-4902
3.4.8	Request for Confirmation of Nomenlature	DI-CMAN-80194
	Request for Assignment of Serial Number and Serial Number Prefix Letters	DI-CMAN-80195
	Request for Approval of Identification Plate Drawings	DI-CMAN-80196
	Request, Assignment of Nomenclature	UDI-E-21582
4.4, 4.6	First Article Inspection Procedure	DI-T-4901
4.4.3.1, 4.4.3.2	Electromagnetic Intereference Test Plan	DI-EMCS-80201
4.4.3.3.c	Report, EMI Test	DI-EMCS-80200
4.5.4.3, 4.6	Procedures, Reliability Tests	DI-RELI-80251
4.5.7.2	Failed Item Analysis Report	DI-RELI-80253
4.7.4	Drawings, Engineering and Associated Lists	DI-E-7031

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed a such in DoD 5010.12L AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 Options. The following options are available:

a. An additional manufacturing screening period is allowable at the option of the contractor (see 4.5.2).

b. Acceptance of equipment may be stopped anytime a reject situation exists in the All Equipment Test. This is at the option of the procuring activity (see 4.5.4.2).

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Associated equipment. The receiver-transmitter should operate with the associated equipment listed in Table XIX which should not be supplied as part of this equipment:

TABLE XIX. Associated equipment

Item	Equipment Designation	Military Specification
Intercommunications Sets	AN/AIC-10, AN/AIC-14 AN/AIC-18, AN/AIC-25 C-6533/ARC	MIL-I-22353 MIL-I-27543 MIL-C-55653
Microphones, Boom Mounted	M-96A/A	MIL-M-22179
Amplifiers, Audio Frequency	AM-3597B/A	MIL-A-23595
Antenna	AS-3191/A AS-3238/A AS-3584/A AS-3585/A	MIL-A-85670
Switch	SA-521A/A with Selector Antenna C-2193B/A	MIL-S-25879
Direction Finder Group	AN/ARC-25, AN/ARC-48 AN/ARA-50, AN/ARD-6987	MIL-D-7030 MIL-D-38402
Speech Security Equipment	TSEC/KY-28 TSEC/KY-58	CSEEB-13 CSEEB-32/ CSESD-14
Radio Set, Survival	AN/PRC-90	MIL-R-81493

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TABLE XIX. Associated equipment (Continued)

Item	Equipment Designation	Military Specification
ANVIS Goggles	AN/AVS-6	ADS-23
Indicator, Frequency/Channel	ID-2121A/ARC	MIL-I-85671
	ID-2229A/ARC	MIL-I-85671
	ID-2303/ARC	MIL-I-85671
Control, Radio Set	C-10319A/ARC	MIL-C-85666
	C-10776A/ARC	MIL-C-85666
	C-11131/ARC	MIL-C-85666
Control, Switching Unit	C-11628/ARC	MIL-C-85674
	C-10777A/ARC	MIL-C-85674
	C-11132/ARC	MIL-C-85674
Control Radio Set	C-11128/ARC	MIL-C-85806
	C-11129/ARC	MIL-C-85806
	C-11130/ARC	MIL-C-85806
Switching Unit	SA-2498/ARC	MIL-S-85673

6.6 Subject term (key word) listing.

AN/ARC-182(V)
Audio Power Output
Center Frequency
Harmonics
HAVE QUICK
Intermodulation
Panel Mounted Receiver-Transmitter
RT-1324A/ARC
Sensitivity
Ultra-High Frequency
Very High Frequency
Voltage Standing Wave Ratio
Transmit Blanking

Preparing Activity:
Navy AS
(Project 5821 - N200)

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